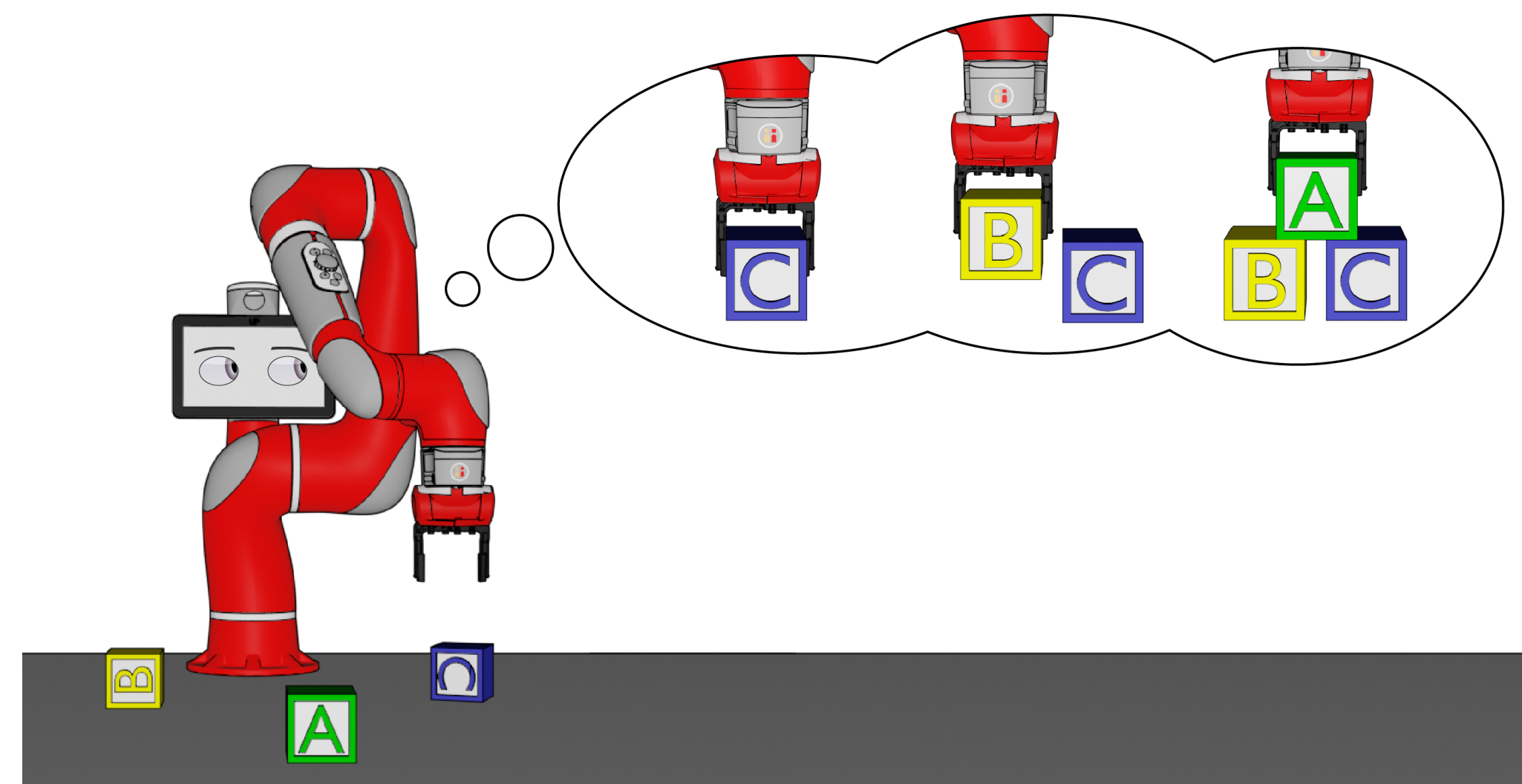


Reinforcement Learning

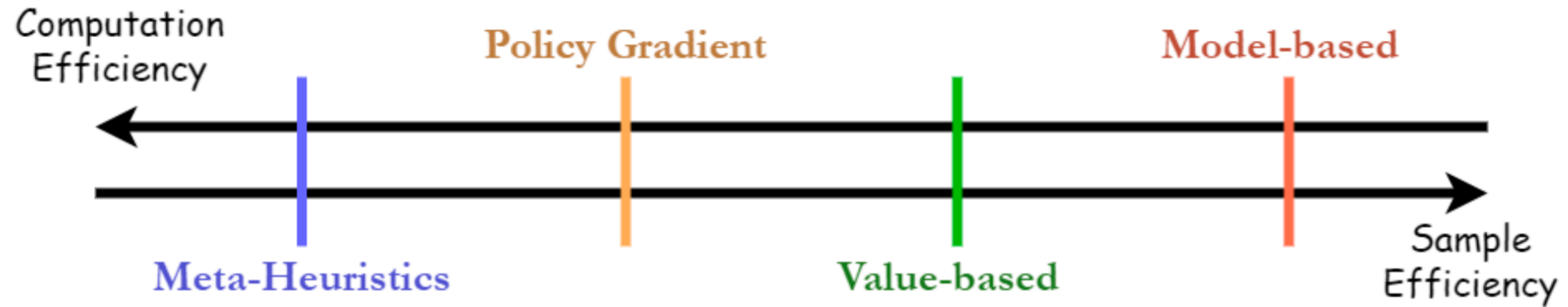
HSE, winter - spring 2026

Lecture 9: Model-based RL

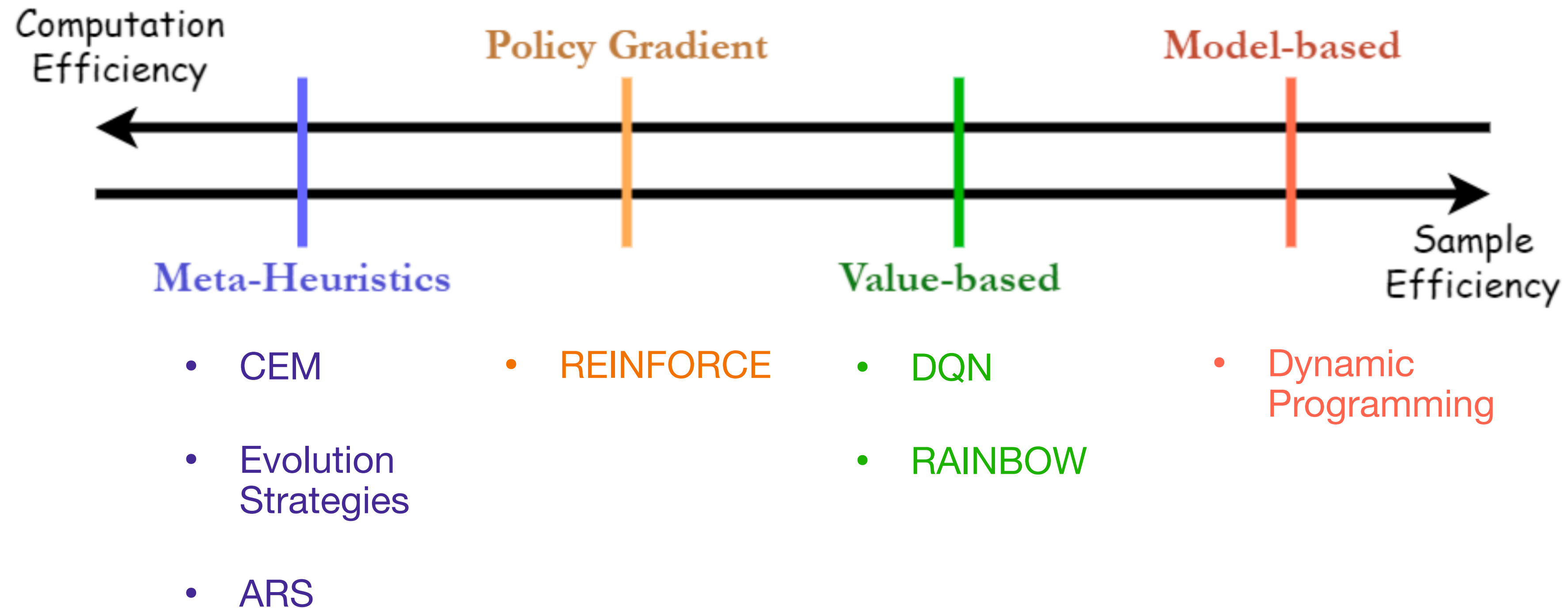


Sergei Laktionov

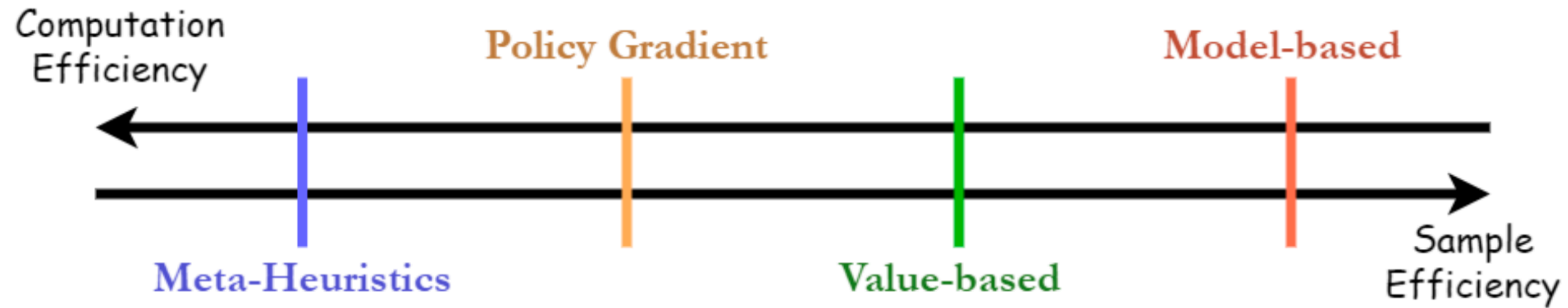
Recap



Recap



Recap



- CEM
- Evolution Strategies
- ARS

- REINFORCE

- A2C
- PPO
- SAC

- DQN
- RAINBOW

- Dynamic Programming

Model-based Setup

The environment's dynamic is known or learnt:

- Stochastic case: $p(r, s' | s, a)$
- Deterministic case: $s' = S(s, a), r = R(s, a)$

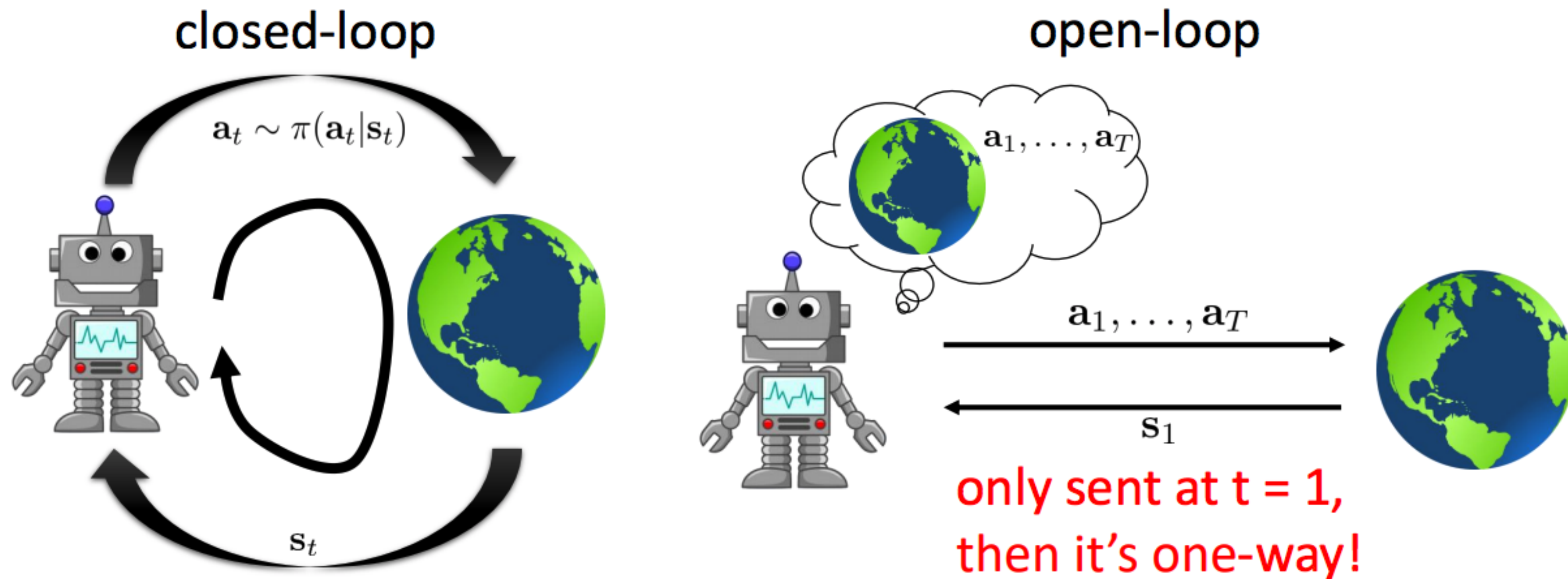
Examples:

- Games (Chess, Go, Shogi etc)

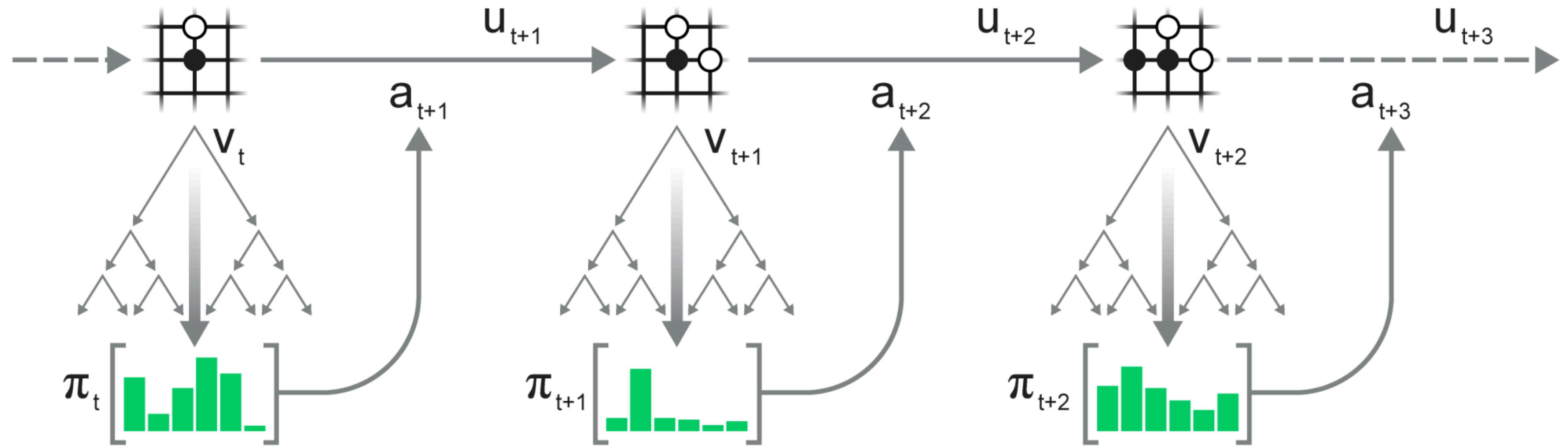
Further in this lecture, we will only consider deterministic environments.

Planning

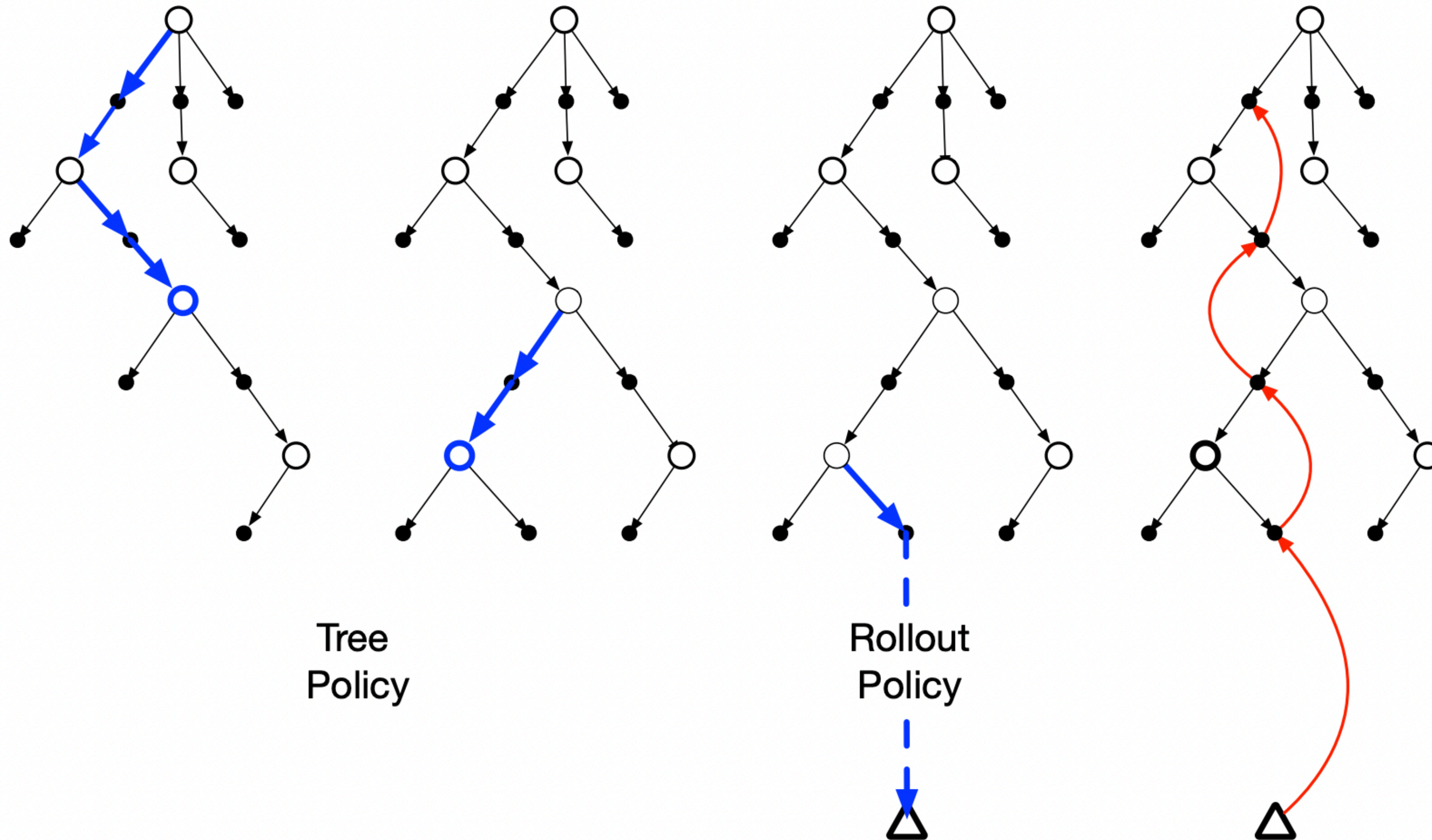
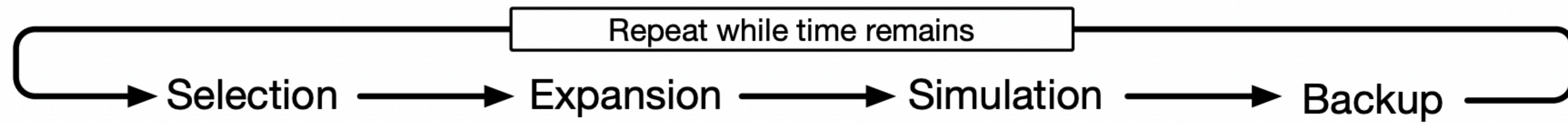
Any computational process that takes an environment as input and produces or improves a policy for interacting with the modelled environment:



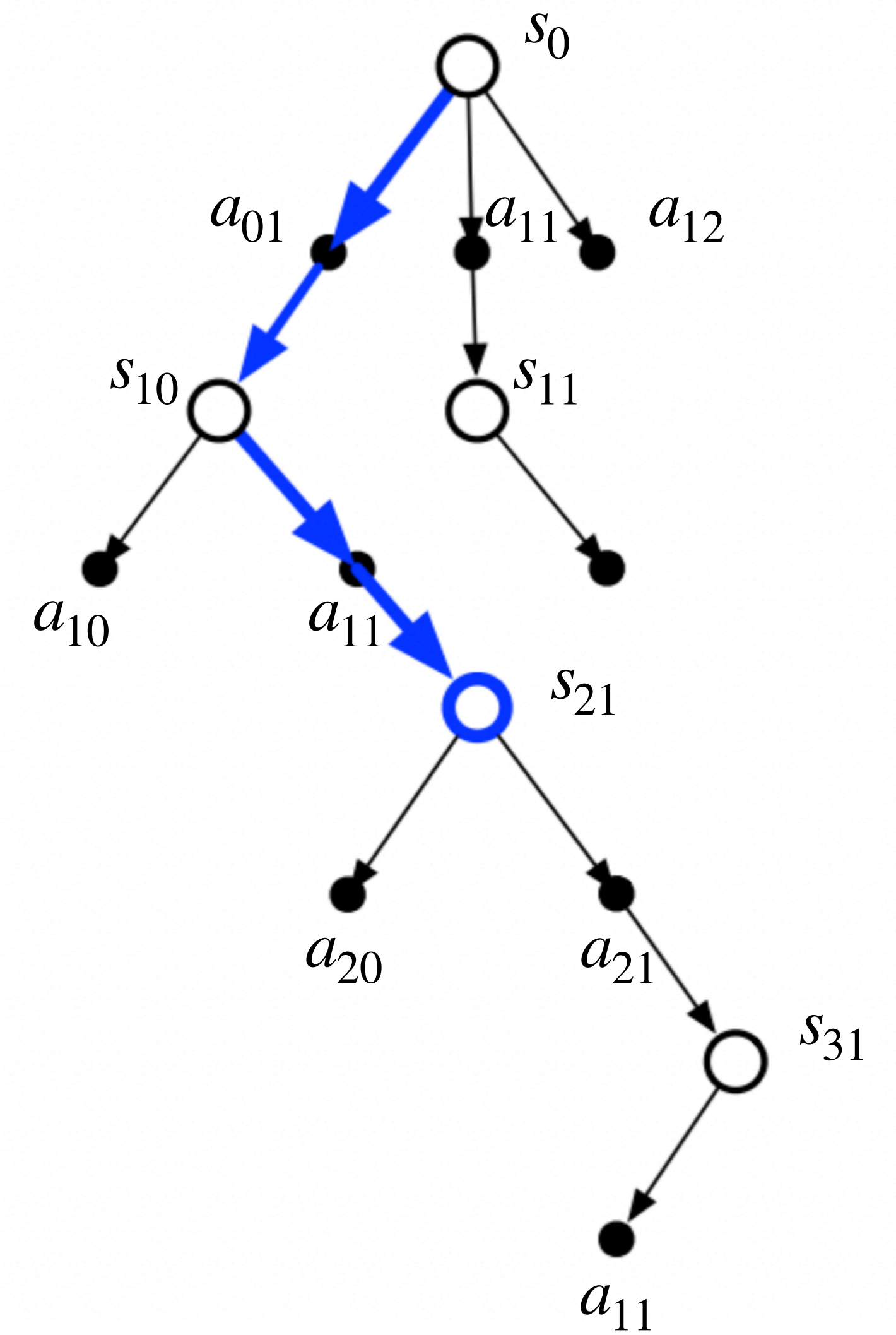
Planning in the Environment



Monte-Carlo Tree Search



Monte-Carlo Tree Search



Monte-Carlo Tree Search

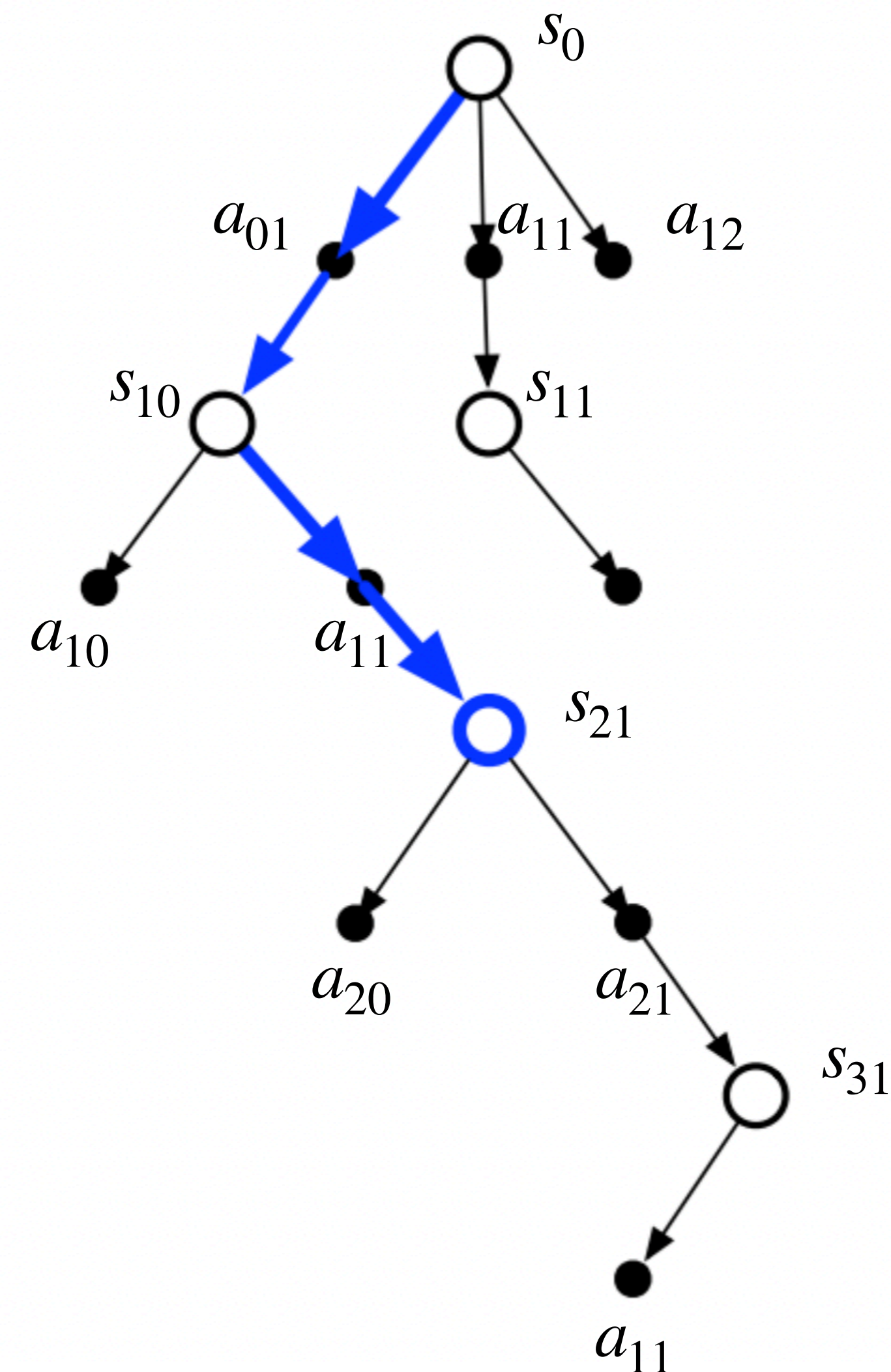
Each node storages a tuple $(G(s, a), n(s, a))$.

Selection:

- Starting from the root, walk down the tree until the leaf node is achieved
- UCB:

$$n(s) = \sum_a n(s, a)$$

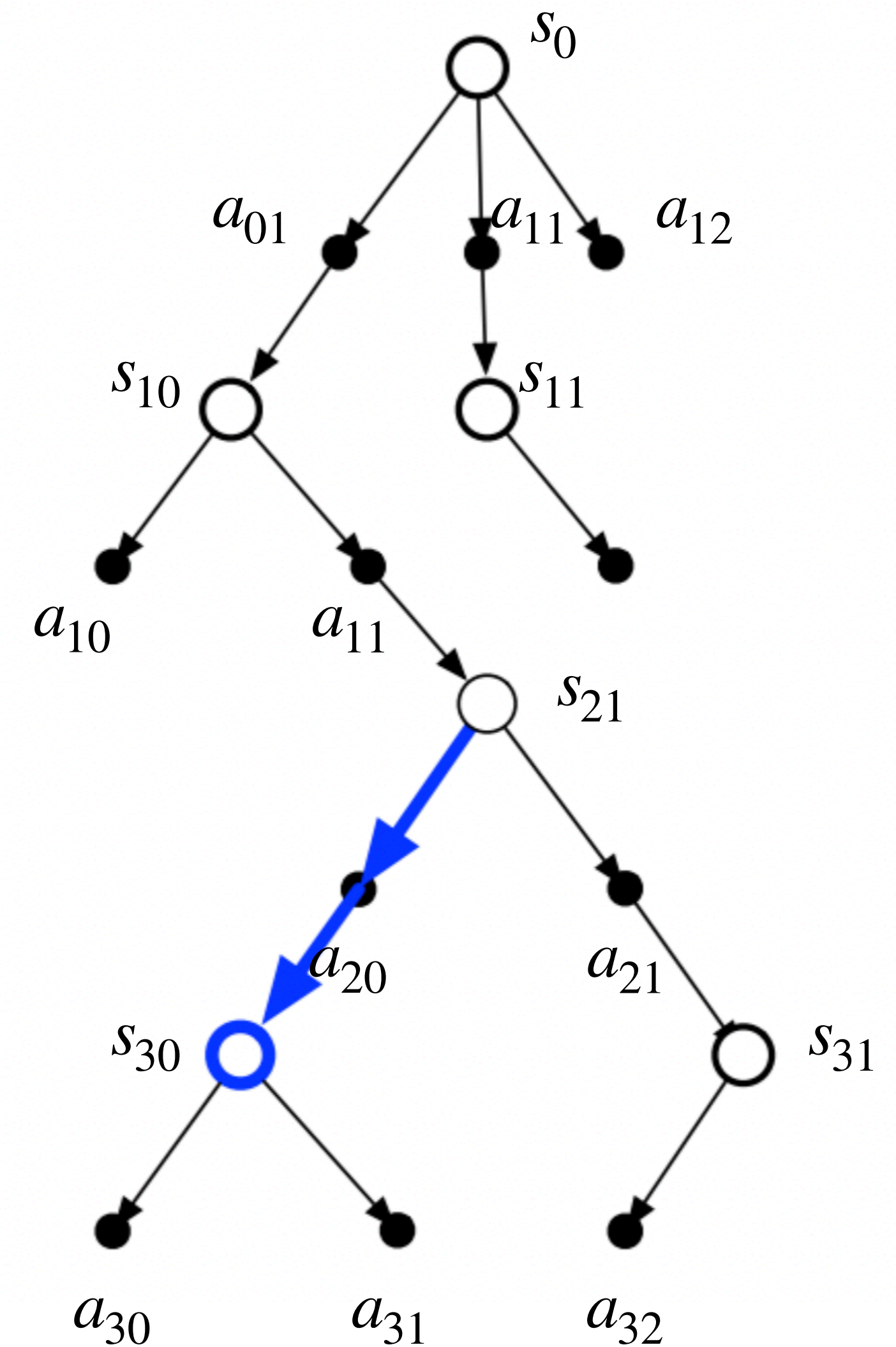
$$\hat{a} = \operatorname{argmax}_a [Q(s, a) + C \sqrt{\frac{\log n(s)}{n(s, a)}}]$$



Monte-Carlo Tree Search

Expansion:

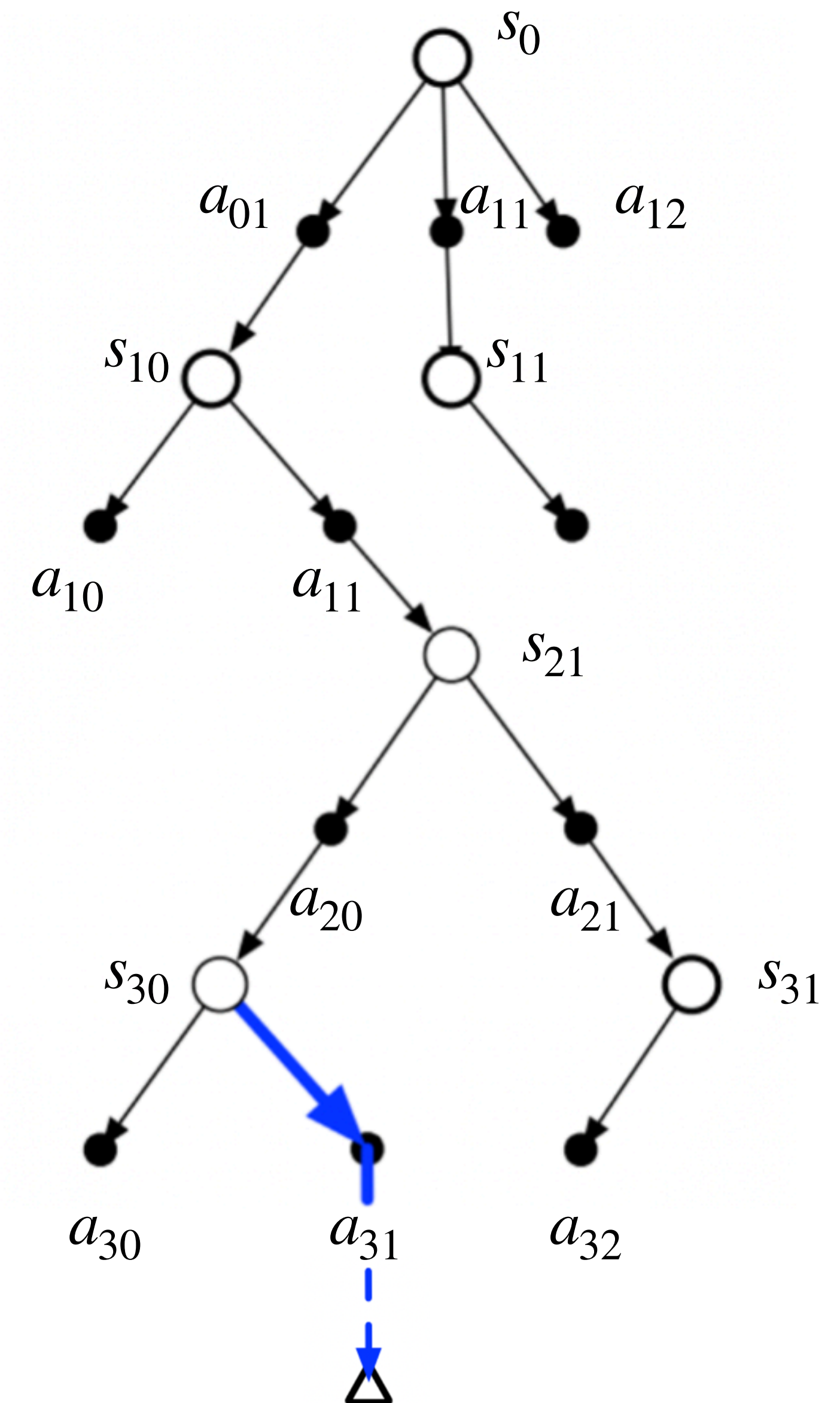
- Select an action a to apply in the state s , either randomly or using a heuristic.



Monte-Carlo Tree Search

Simulation:

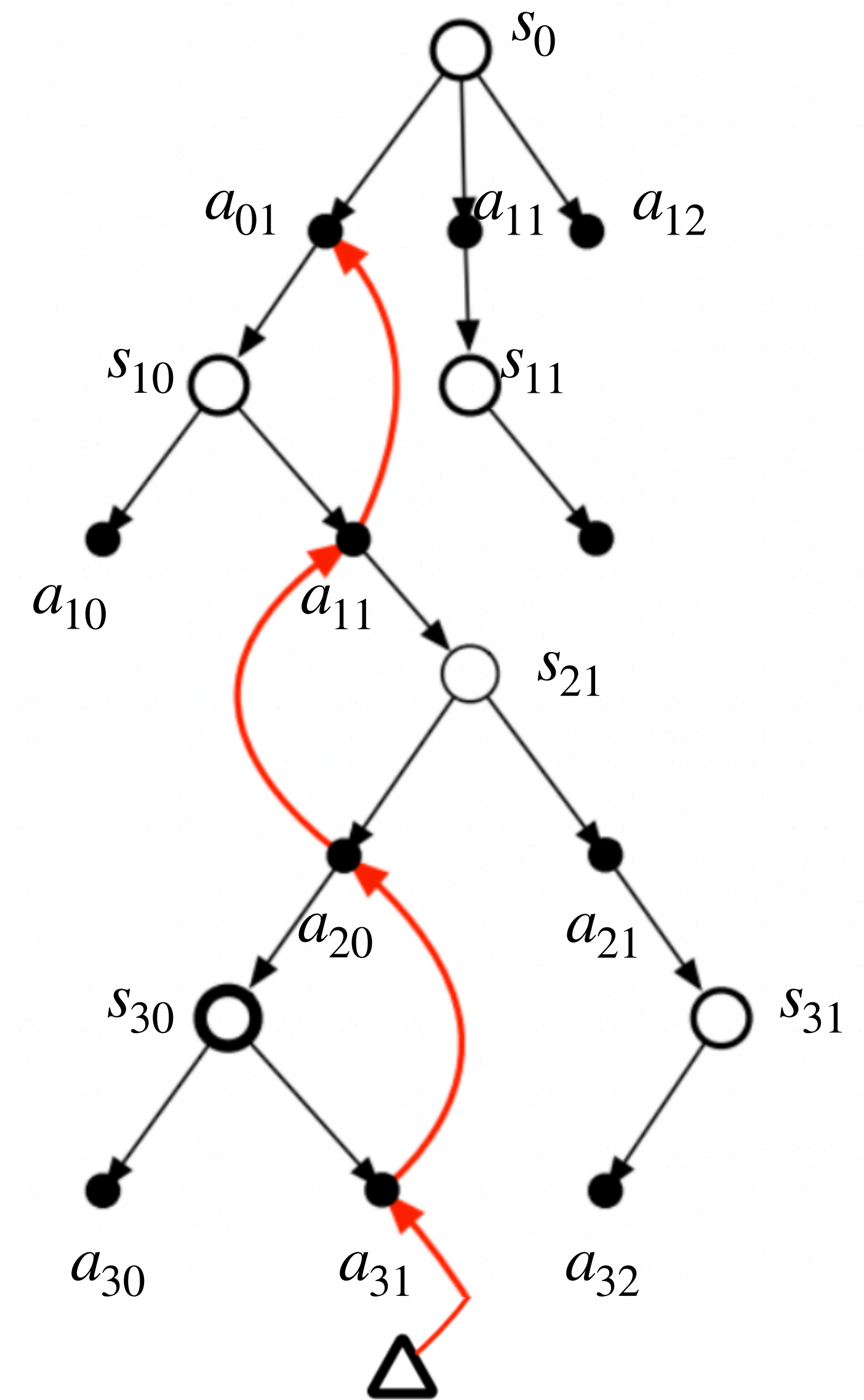
- Perform a randomised simulation of the MDP until we reach a terminating state.



Monte-Carlo Tree Search

Propagation:

- $n(s, a) \leftarrow n(s, a) + 1$
- $G \leftarrow r + \gamma G$
- $Q(s, a) \leftarrow Q(s, a) + \frac{1}{n(s, a)}[G - Q(s, a)]$



Monte-Carlo Tree Search

The result of the MCTS algorithm is a policy in the current environment's state:

- $\pi_{MCTS}(a | s) \propto n(s, a)^{\frac{1}{\tau}}$
- $\pi_{MCTS}(s) = \operatorname{argmax}_a Q(s, a)$

Self-Play

- The zero-sum game of two players
- $o = \begin{cases} 1, & \text{if now it's first player's turn} \\ -1, & \text{otherwise} \end{cases}$
- $Q_o(s, a) = oQ(s, a)$
- $\hat{a} = \operatorname{argmax}_a [Q_o(s, a) + C\sqrt{\frac{\log n(s)}{n(s, a)}}]$
- $\pi_{MCTS}(a \mid s) \propto n(s, a)^{\frac{1}{\tau}}$

MCTS

Two issues:

1. After applying an action sample from $\pi_{MCTS}(\cdot | s)$ we probably remove the tree's nodes corresponding to not chosen actions.
2. We must play till the end to get an estimation of $V(s)$.

MCTS

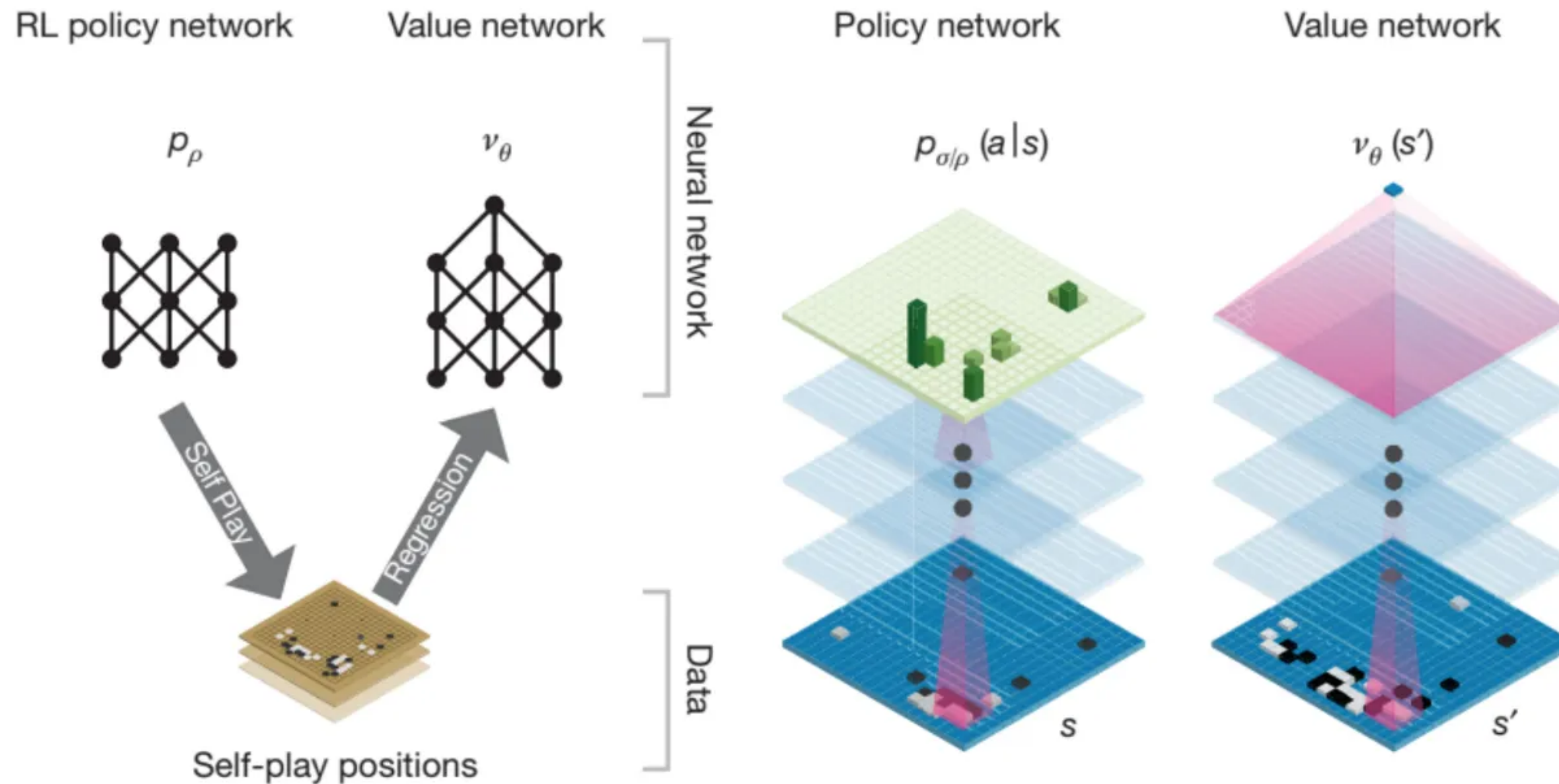
Two issues:

1. After applying an action sample from $\pi_{MCTS}(\cdot | s)$ we probably remove the tree's nodes corresponding to not chosen actions.
2. We must play till the end to get an estimation of $V(s)$.

Two ideas:

1. Distill π_{MCTS} in p_θ .
2. $V \approx V_\phi$

Architecture

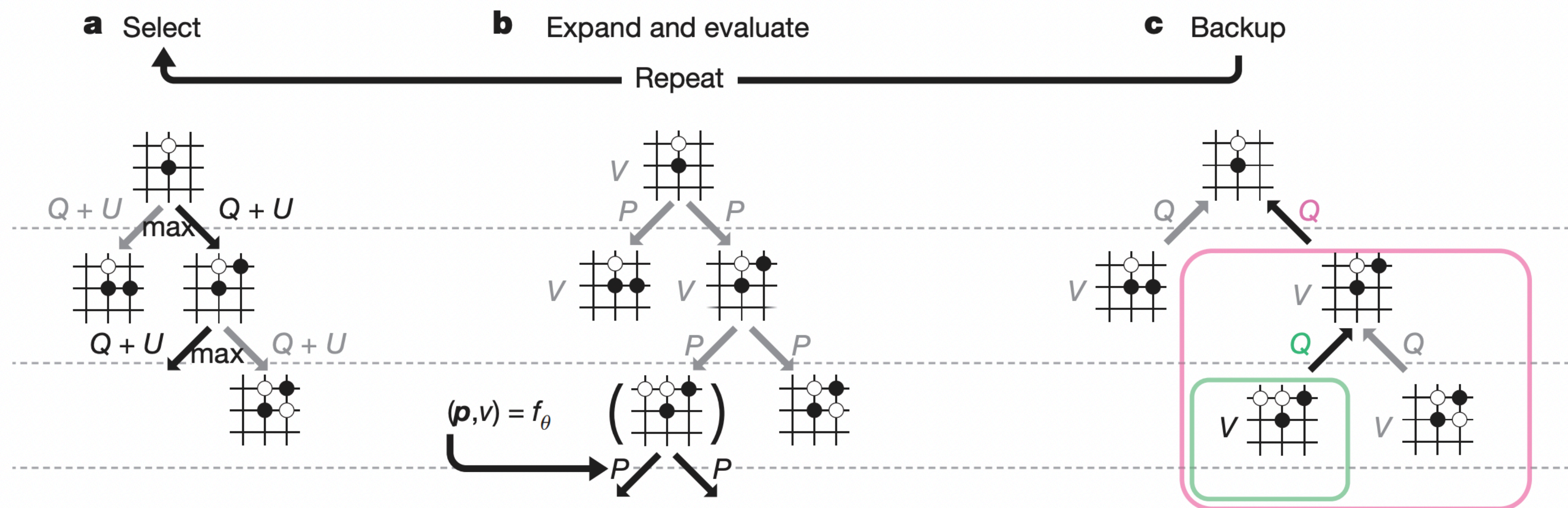


Policy and Value networks are indeed combined

MCTS in AlphaZero

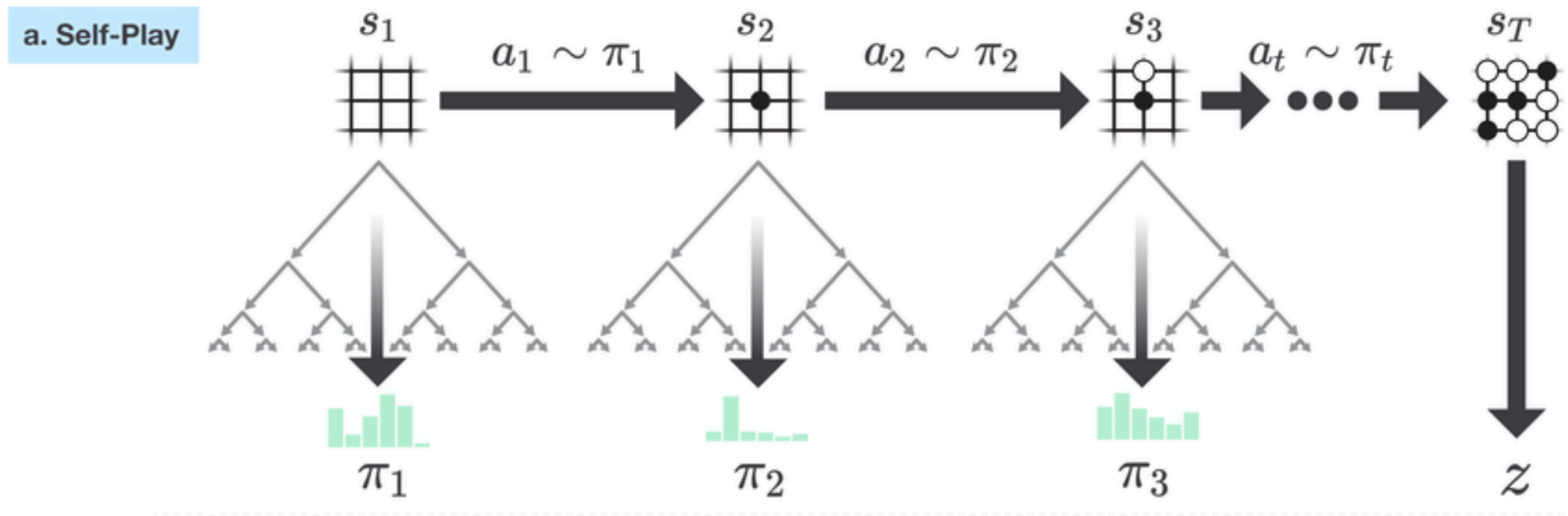
$f_{\theta}(s) = (\mathbf{p}, V)$ is a two-head neural network.

1. Selection: $\hat{a} = \operatorname{argmax}_a [\hat{Q}(s, a) + C p_{\theta}(a | s) \frac{\sqrt{n(s)}}{n(s, a) + 1}]$
2. Expansion
3. Propagation: for leaf state s_{leaf} use $V_{\theta}(s_{leaf})$



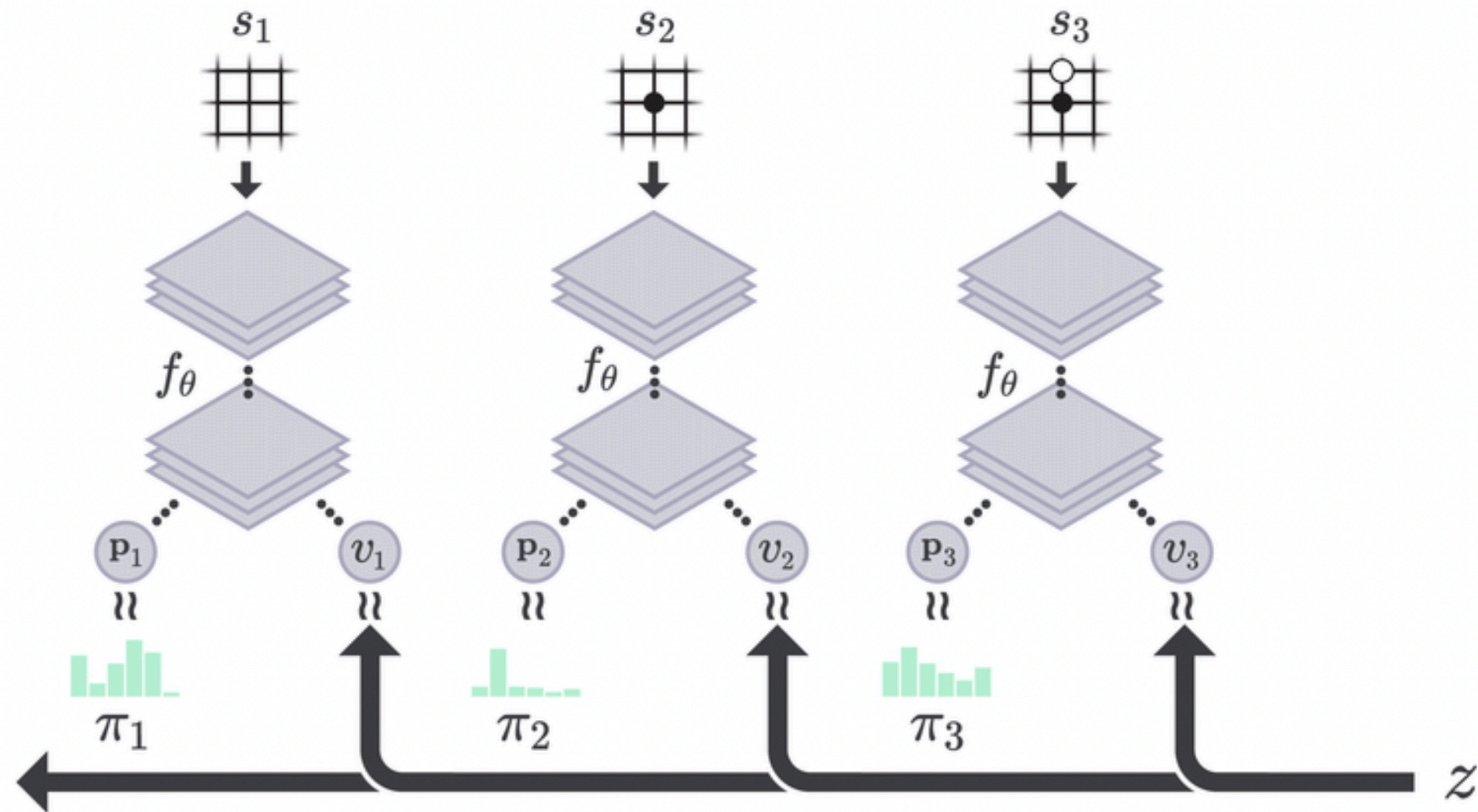
AlphaZero Network Training

1. Many self-play games: in each state, an agent plans with MCTS and executes action $a \sim \pi(\cdot | s) \propto n(s, a)^{\frac{1}{\tau}}$
2. Store the dataset of triplets $(s, \pi_{MCTS}(\cdot | s), z)$.



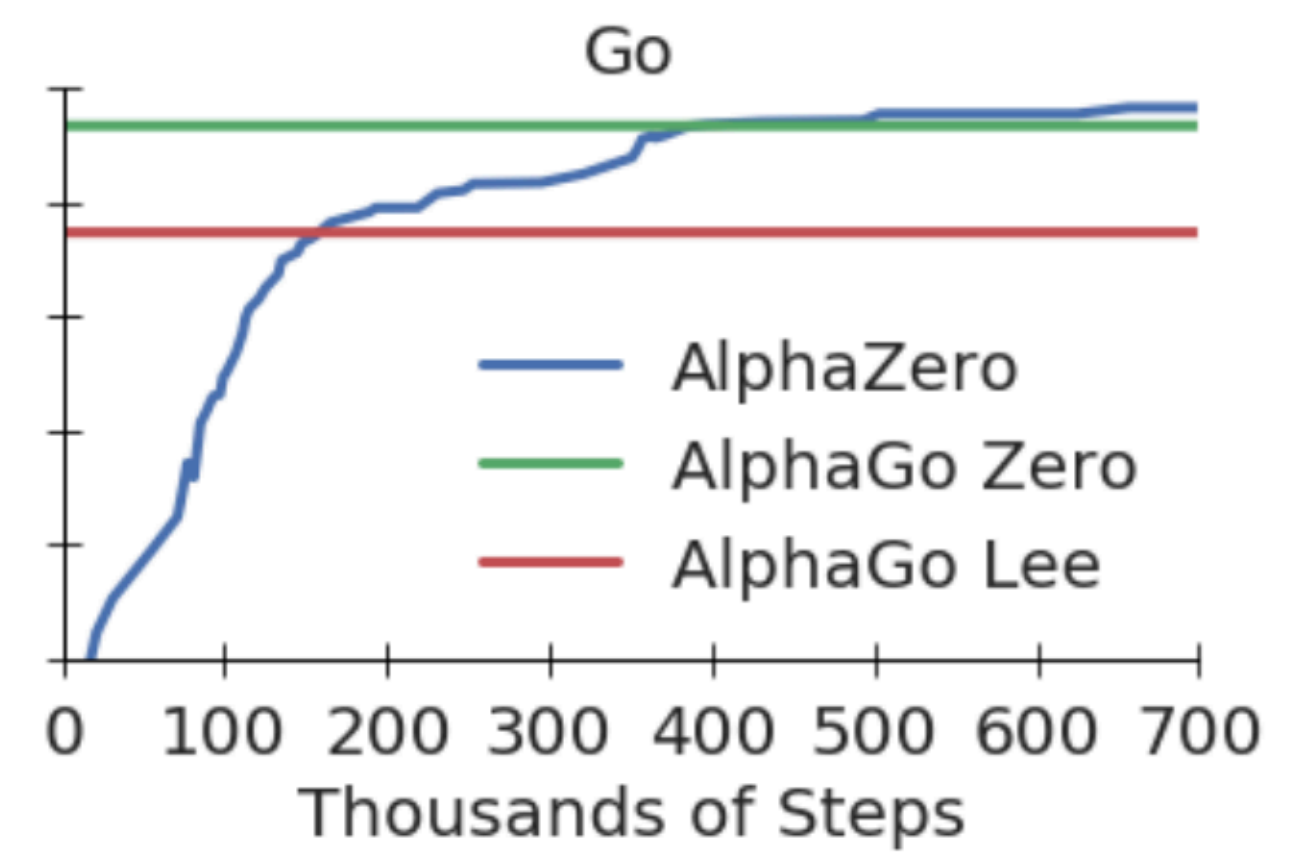
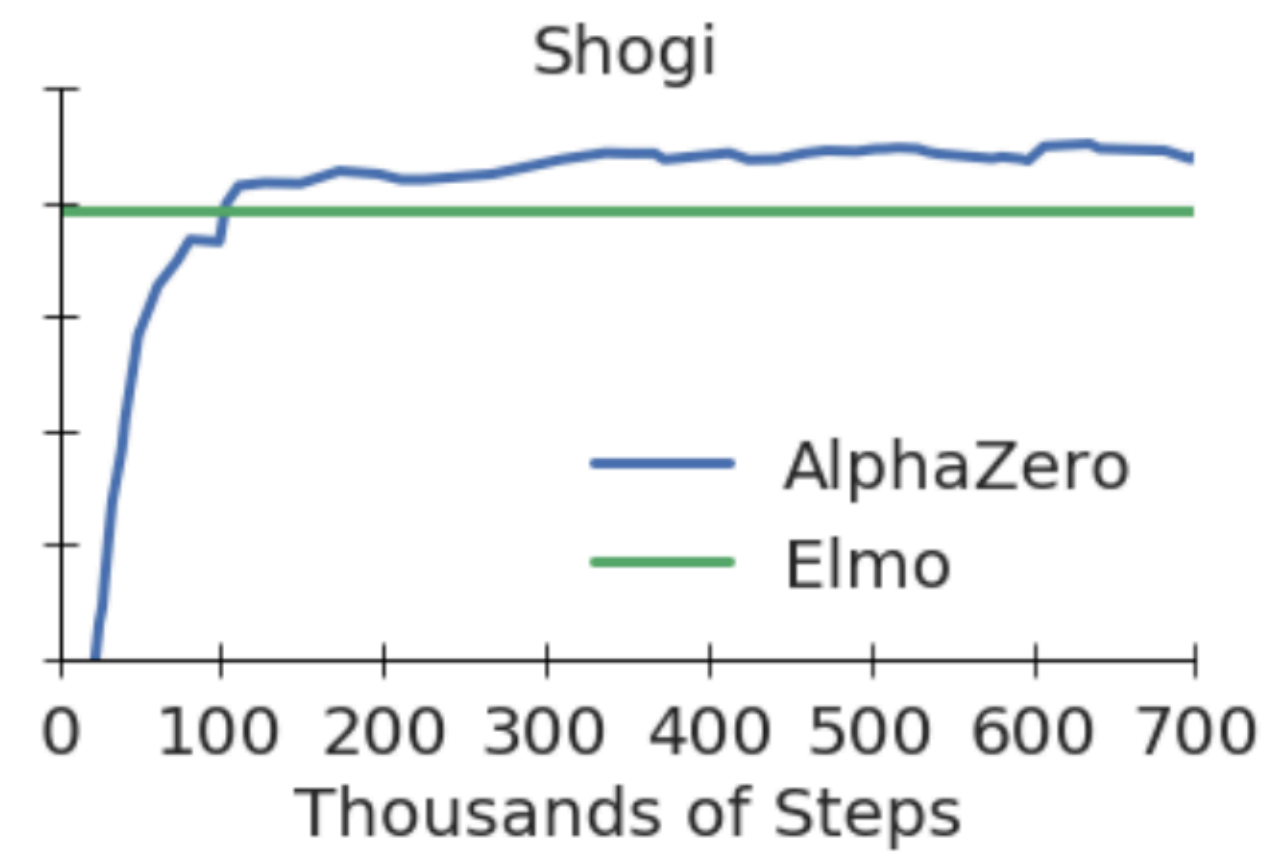
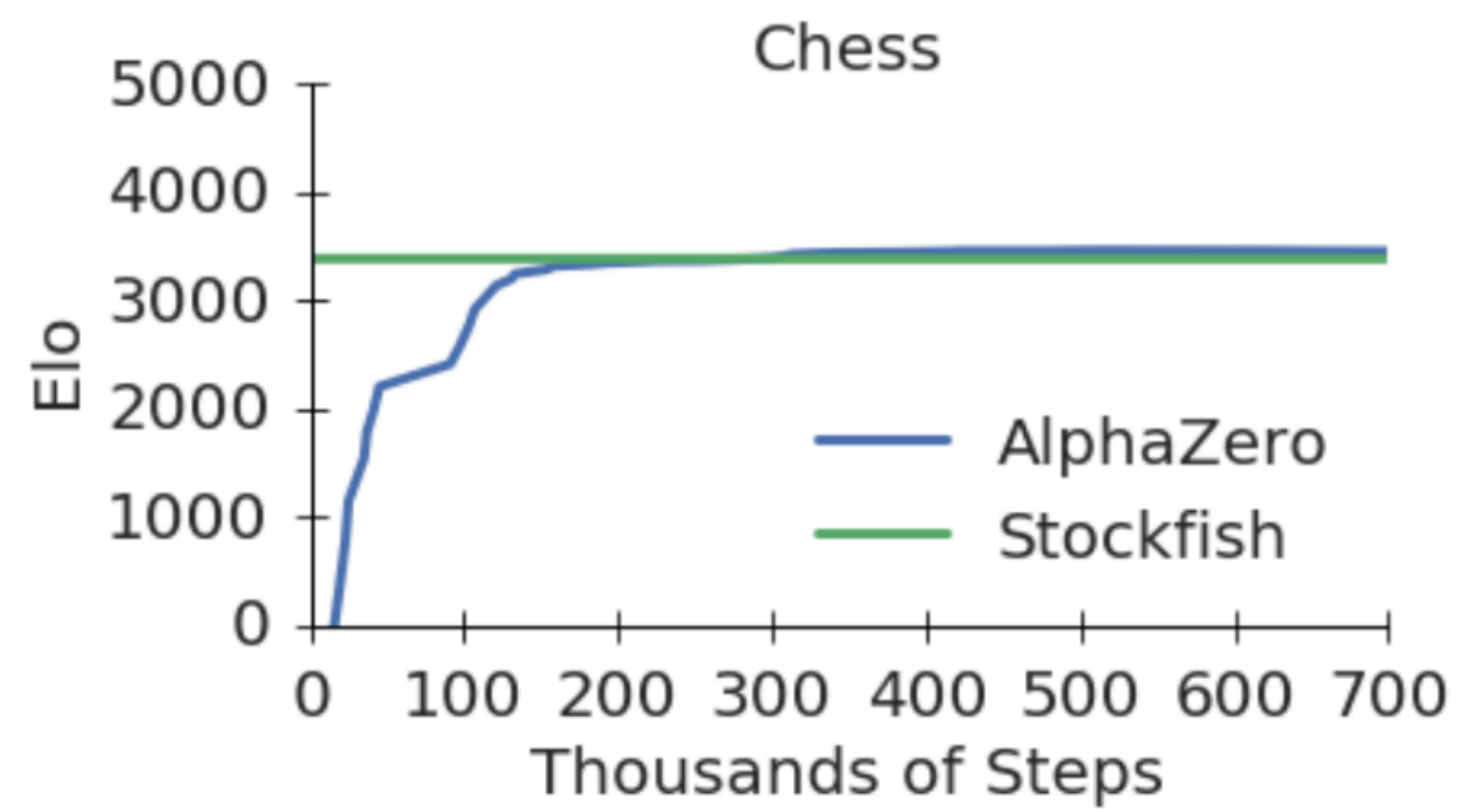
AlphaZero Network Training

b. Neural Network Training



$$(z - V_\theta(s))^2 + \sum_a \pi_{MCTS}(s | a) \log p_\theta(a | s) + c ||\theta||_2^2 \rightarrow \min_{\theta}$$

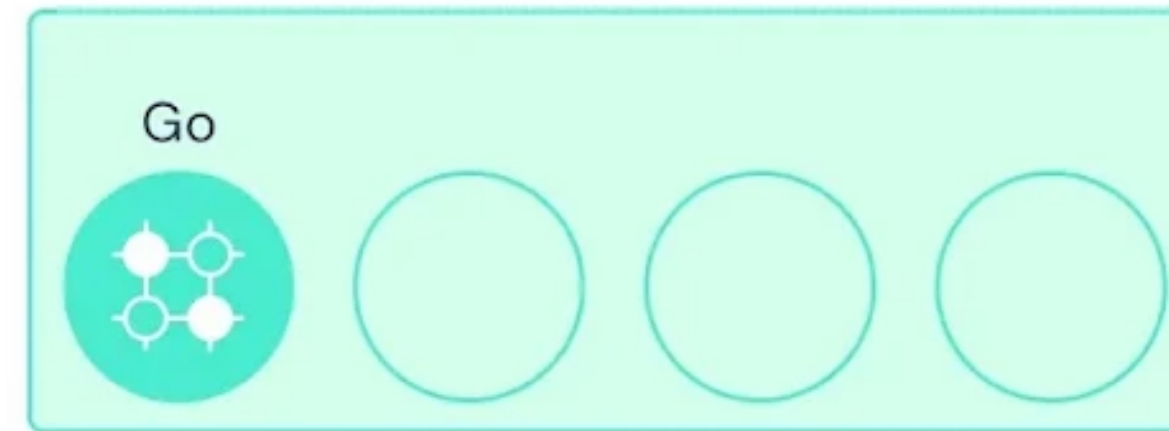
AlphaZero Results



[Link](#)



Domains

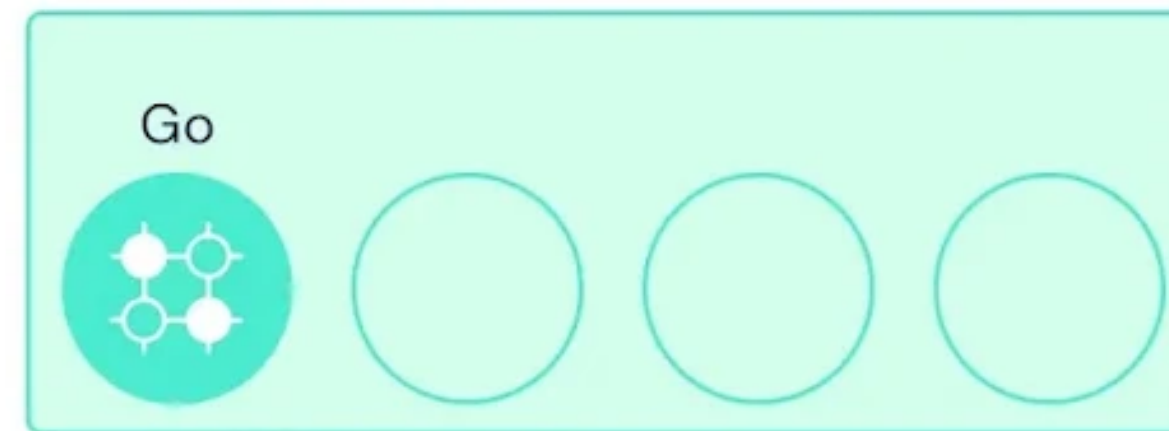


Knowledge



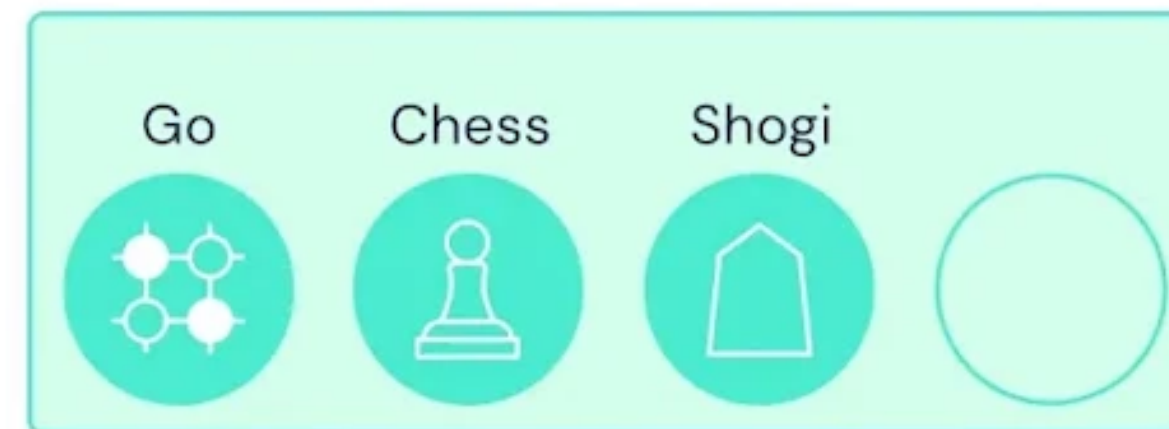
AlphaGo becomes the first program to master Go using neural networks and tree search (Jan 2016, Nature)

[Link](#)



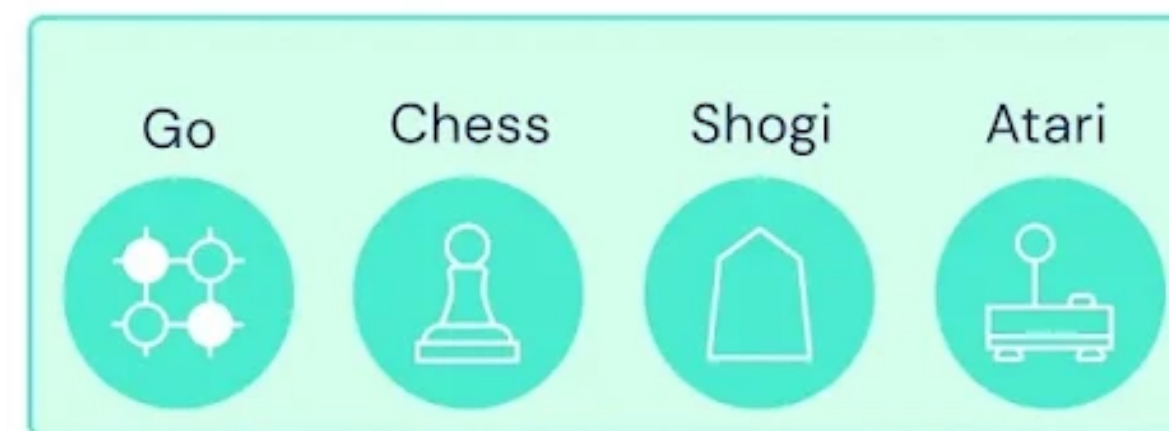
AlphaGo Zero learns to play completely on its own, without human knowledge (Oct 2017, Nature)

[Link](#)



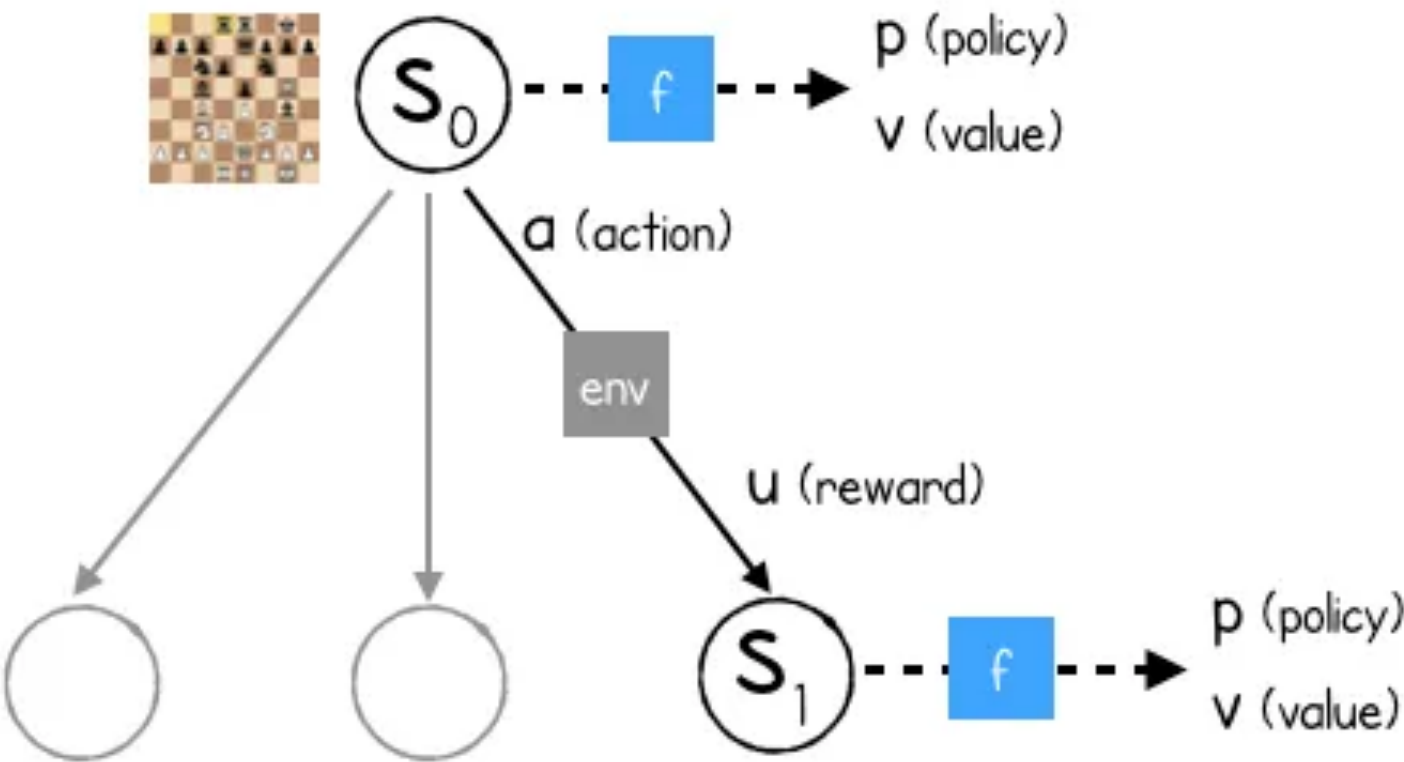
AlphaZero masters three perfect information games using a single algorithm for all games (Dec 2018, Science)

[Link](#)



Comparison

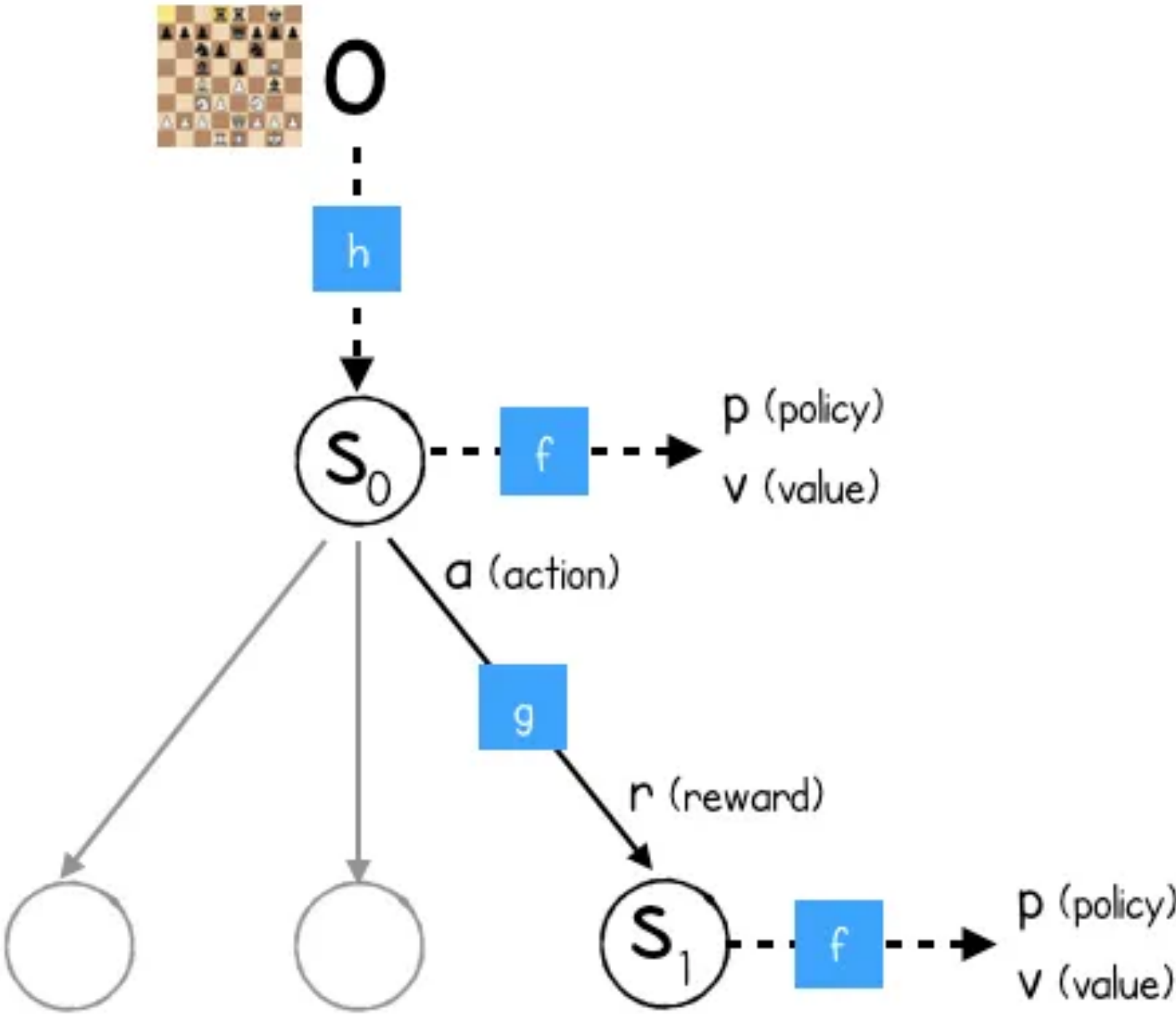
AlphaZero



AlphaZero has 1 network

	from	to
prediction f :	s	p, v

MuZero



MuZero has 3 networks

		from	to
prediction	f :	s	p, v
dynamics	g :	s, a	r, s
representation	h :	o	s

- Representation:

$$h_{\theta}(o_1, \dots, o_t) = s_t^0$$

- Prediction:

$$f_{\theta}(s_t^k) = (p_t^k, v_t^k)$$

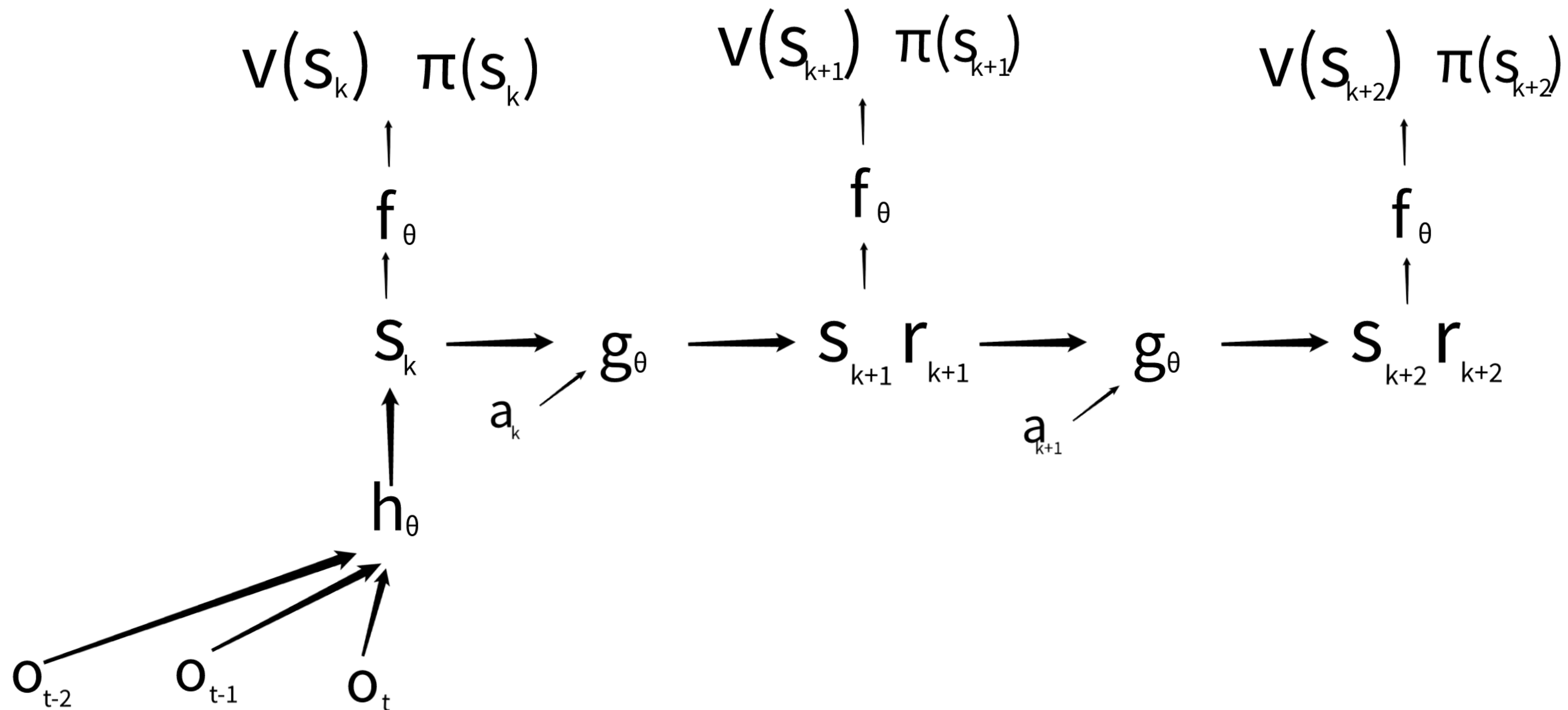
- Dynamics:

$$g_{\theta}(s_t^{k-1}, a_t^k) = (r_t^k, s_t^k)$$

MuZero Loss

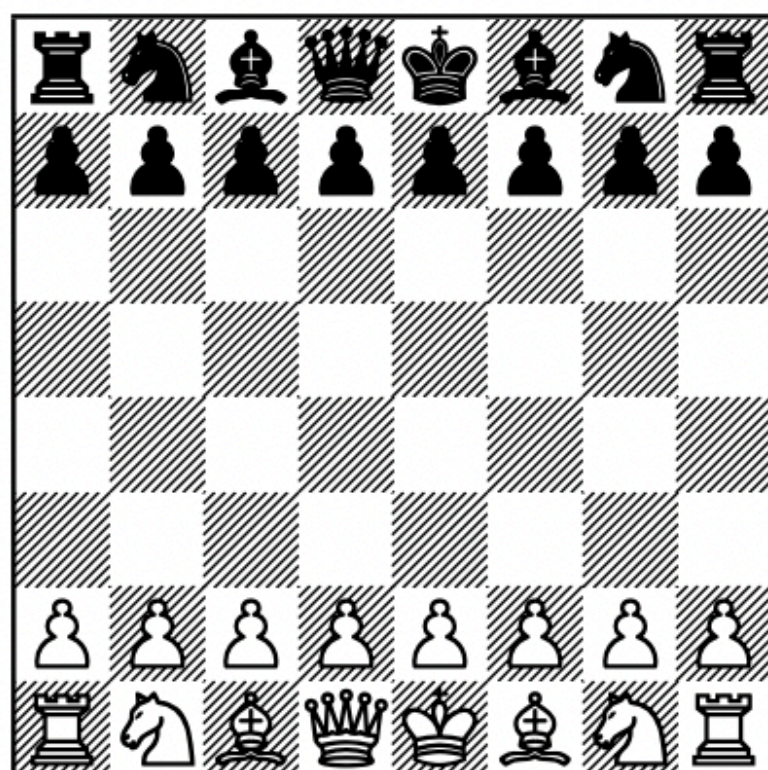
$$l_t(\theta) = \sum_{k=0}^K [(u_{t+k} - r_t^k)^2 + (z_{t+k} - v_t^k)^2 - \pi_{t+k} \log p_t^k] + c ||\theta||_2^2 \rightarrow \min_{\theta}$$

- u_{t+k} is an immediate reward
- z_{t+k} is n -step Bellman target



MuZero Results

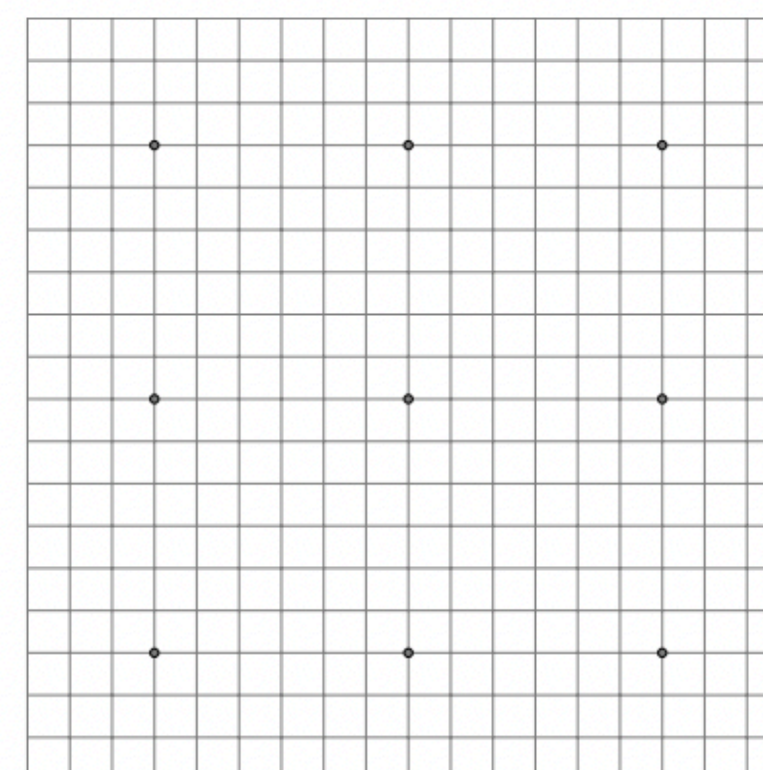
Chess



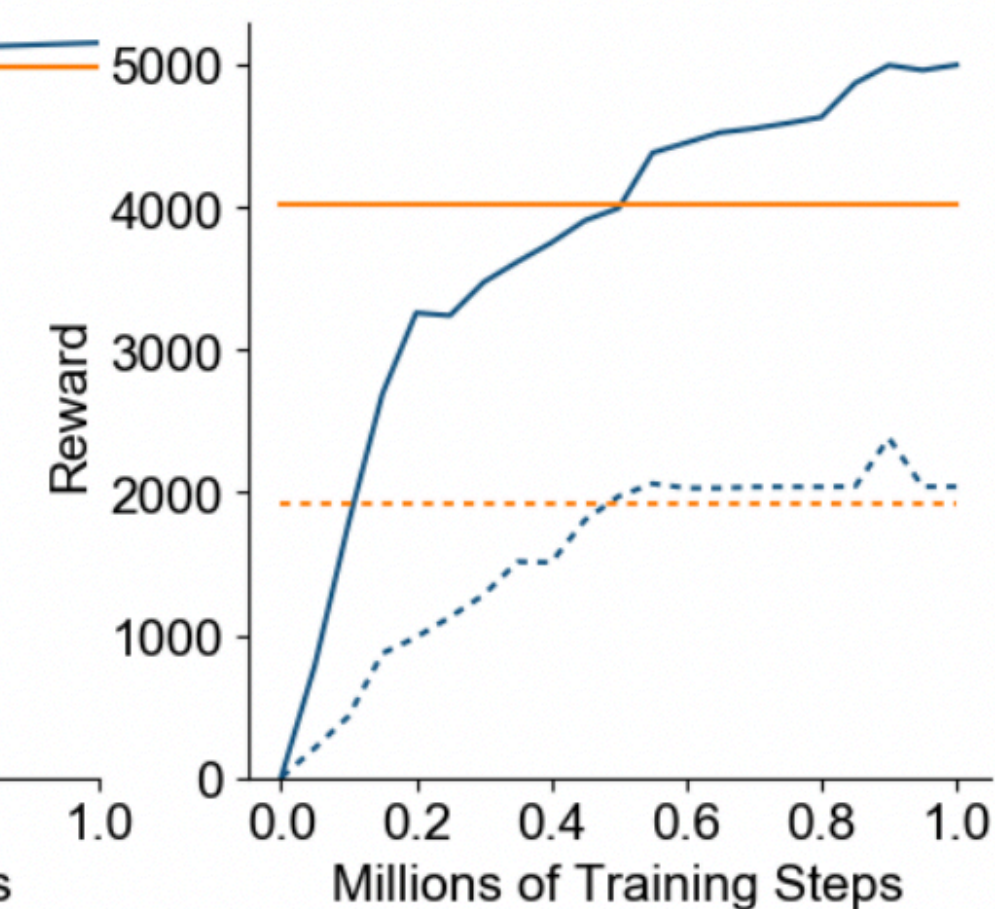
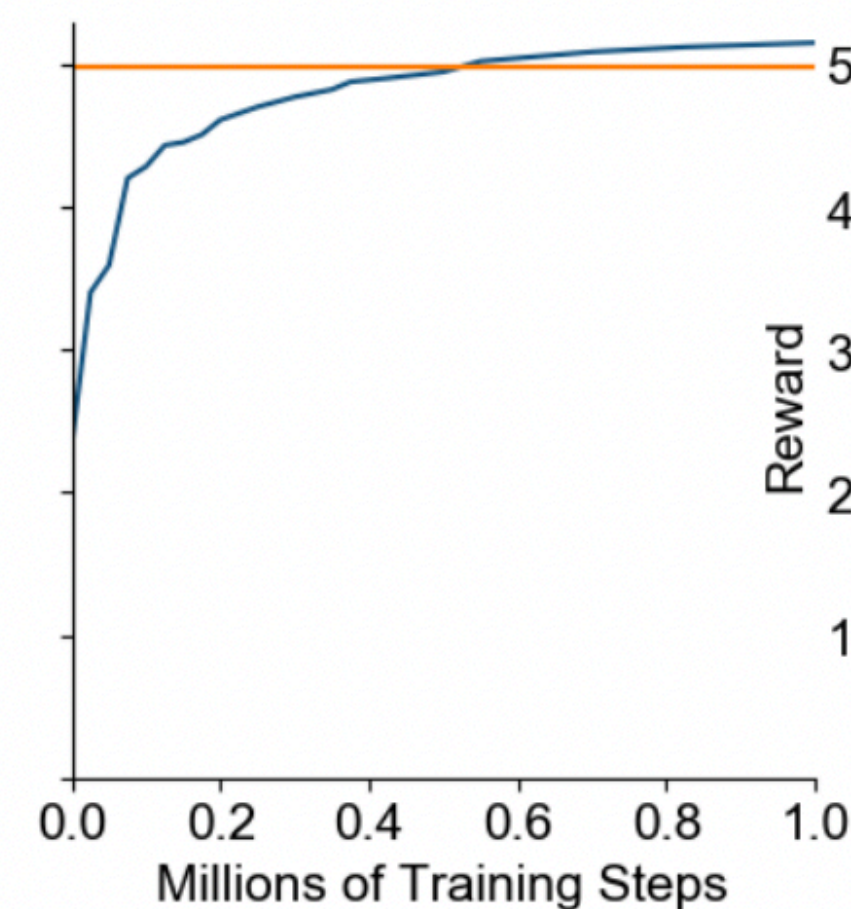
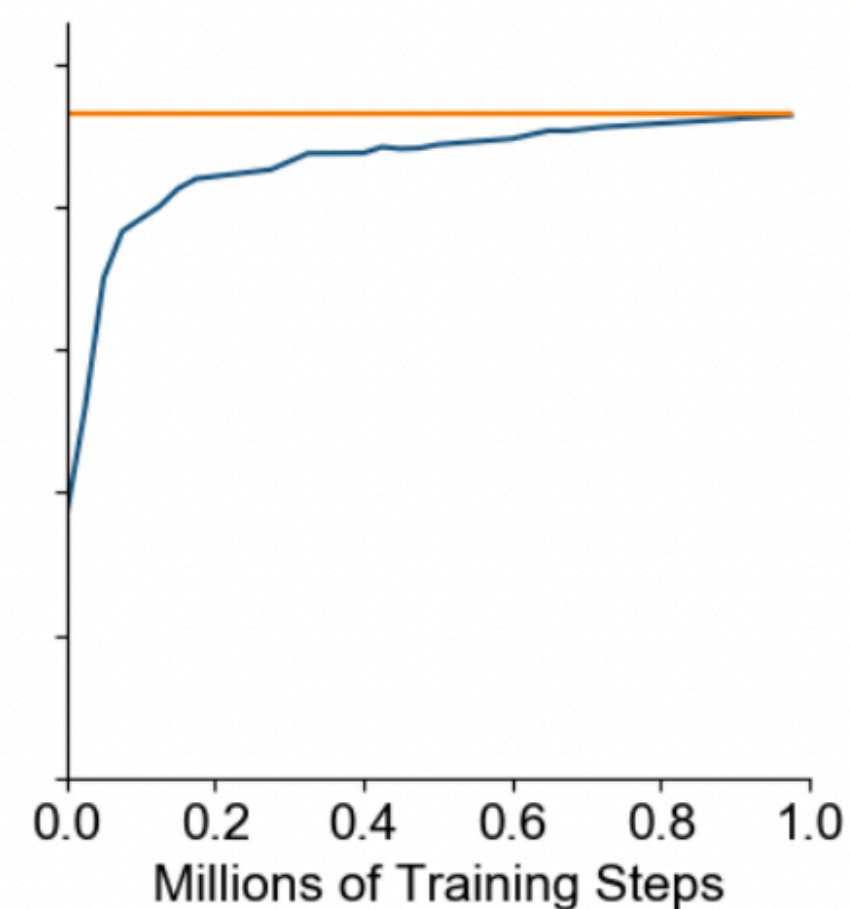
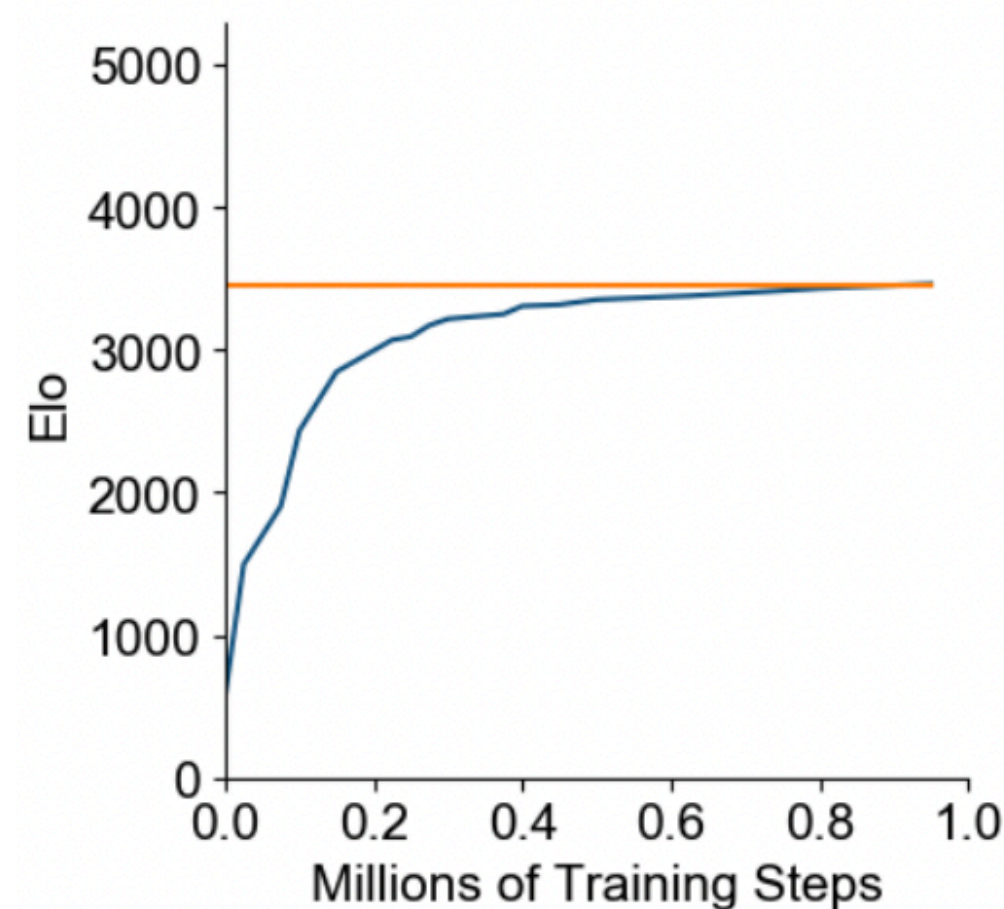
Shogi



Go



Atari



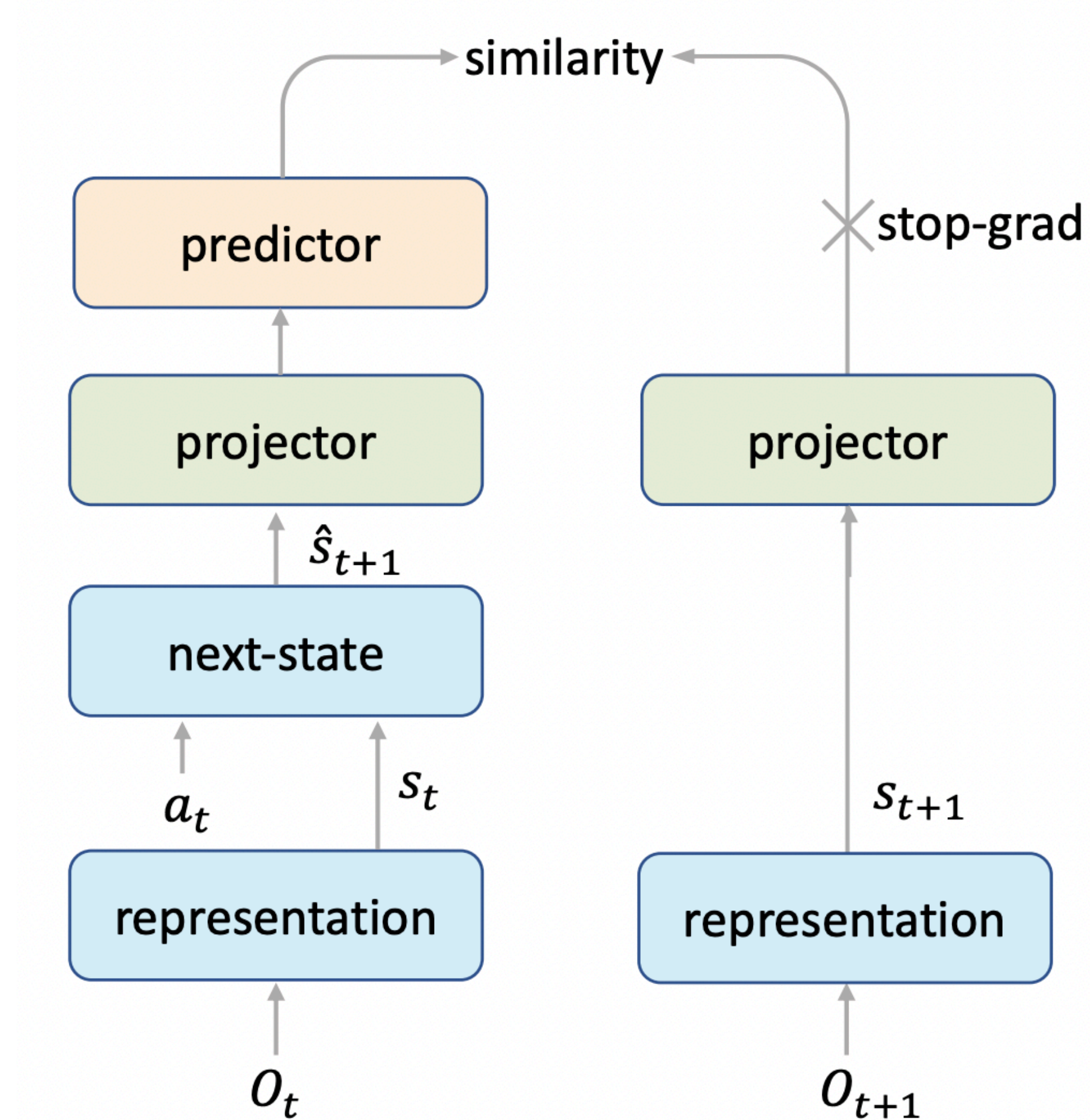
EfficientZero

MuZero issues:

1. Lack of supervision on environment model:
 - The reward can be sparse
 - Targets are calculated with bootstrapping
2. Hardness to deal with aleatoric uncertainty:
 - the predicted rewards still have large prediction errors.
3. Off-policy issues of multi-step value

Self-Supervised Consistency Loss

1. Compare the latent representation of consecutive observations with negative cosine similarity and optimise for it.



Self-Supervised Consistency Loss

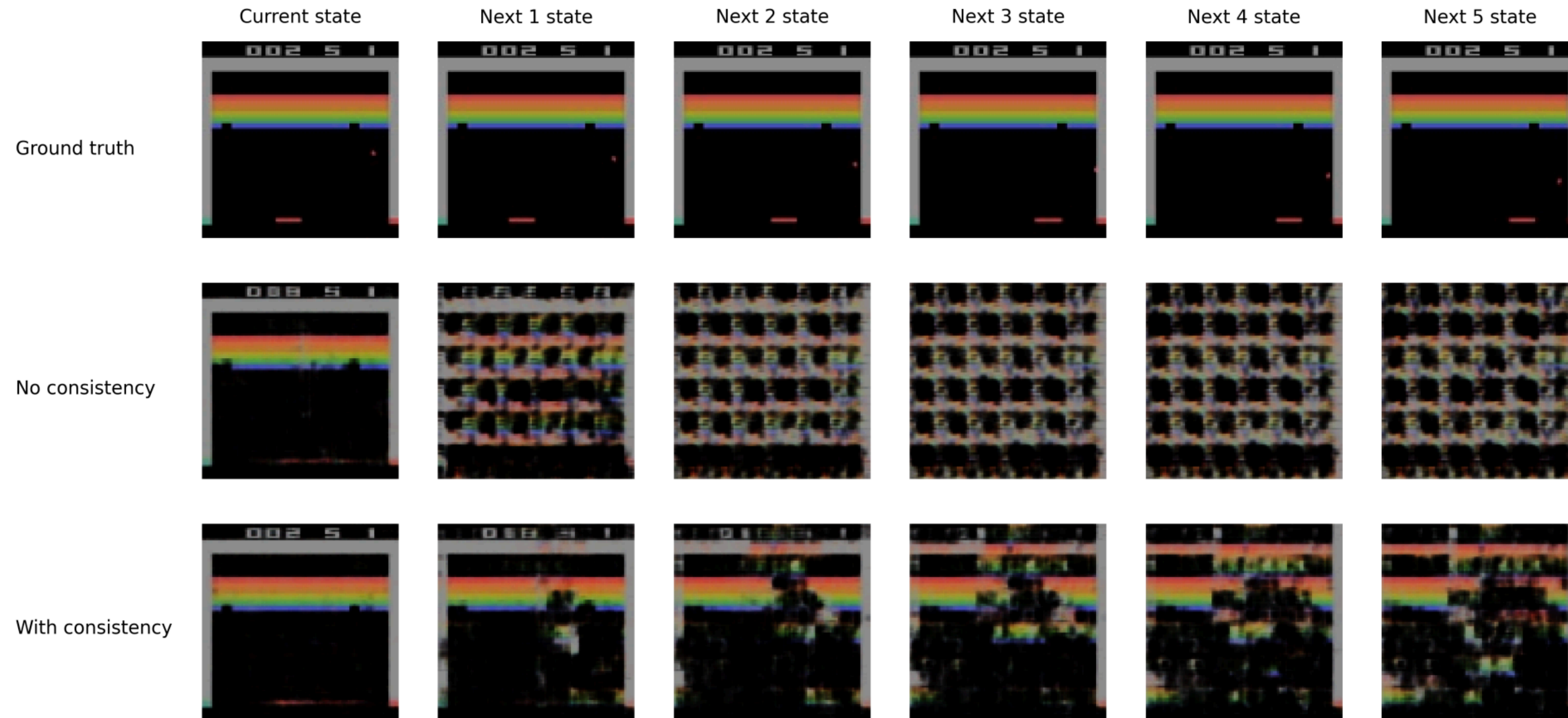


Figure 4: Evaluations of image reconstructions based on latent states extracted from the model with or without self-supervised consistency. The predicted next states with consistency can basically be reconstructed into observations while the ones without consistency cannot.

End-To-End Prediction of the Value Prefix

2. Predict the value prefix, not immediate rewards:

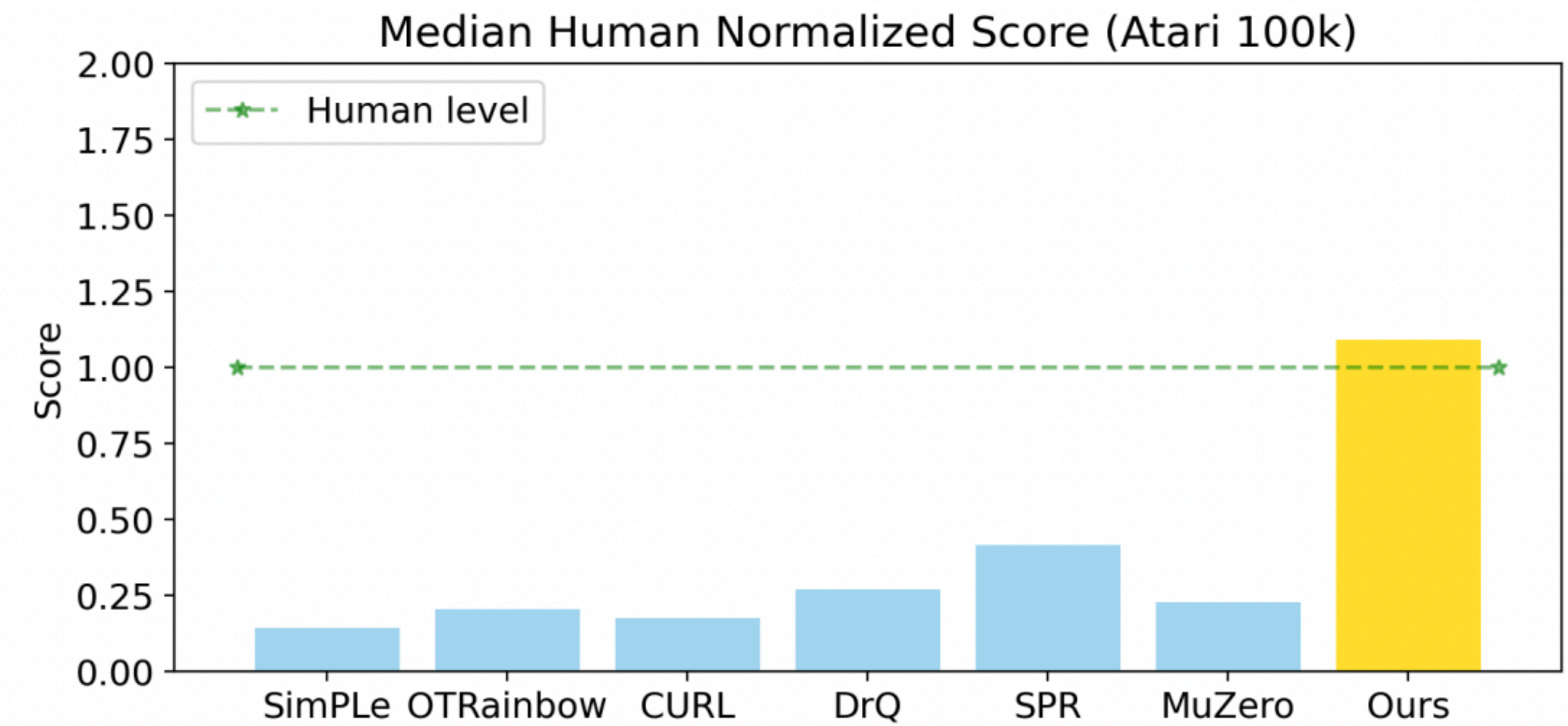
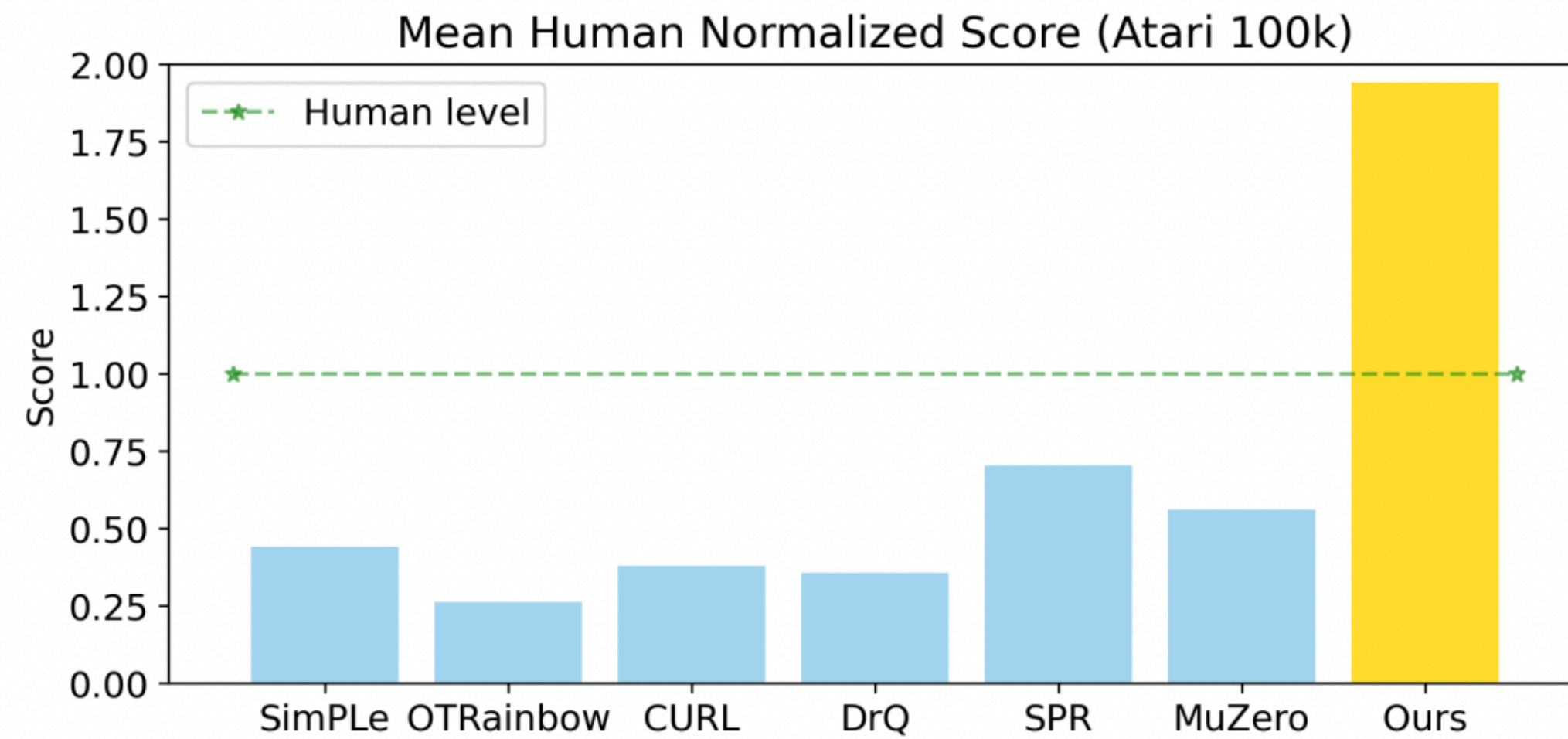
$$Q(s_t, a) = \sum_{k=0}^{K-1} \gamma^k r_{t+k} + \gamma^K V(s_{t+K})$$

Model-Based Off-Policy Correction

3. Use rewards of a dynamic horizon L from the old trajectory and redo an MCTS search with the current policy on the last state s_{t+L} and compute the empirical mean value at the root node.:

$$z_t = \sum_{k=0}^{L-1} \gamma^k u_{t+k} + \gamma^L V^{MCTS}(s_{t+L}), L < K$$

EfficientZero Results



Dreamer-V1



World Model
Learning



Learning Value and
Actor Networks



Environment
Interaction

Learning the World Model



Efficient Behaviour Learning



O_1

Background

1. Practical RL course by YSDA, week 10
2. Reinforcement Learning Textbook (in Russian): 7.2-7.3
3. DreamerV3
4. EfficientZeroV2

Thank you for your attention!